Thread - 2025-11-11

https://x.com/RiikonenMarko/status/1988223318281912802

1/24 - 2025-11-11 12:32 UTC

10/11 Nov 2025, Rovaniemi. The first glitter. Left apartment yesterday around 7 pm and was back today around 2pm. Just these two photos now to give an idea of the action. Temps were perfect, it started column at midnight and ended plate at the daybreak. Solar could have been good, too, but every time I arrived to the scene, action was already dwindling.

I spent all the night's action hours (some 5 in total) on the highest bridge at the ski center. Lot's of light pollution, they had all the slope lights on, but I wanted to do extreme low elevation lamp.

2/24 - 2025-11-12 07:38 UTC

10/11 Nov 2025, Rovaniemi. After I had stopped photographing on the bridge due to the lights going on all around at 6 am for cross-country skiers, I took some documentation of streetlight halos. At this point it was heavily plate dominated, giving quite nice parhelia. At the beginning of the video a truck passes by, causing wave go through the stripes. Too bad camera was trying to focus on my hand.

All halos are of course affected, and in earlier chases we have seen also pillars getting disturbed by passing trucks. Gusts of wind combined with suitable topography (such as if you are on top of an earth mound) can cause similar effects.

[video: 1988511785373106374-WDcrbWzCCEEh7jzi.mp4]

3/24 - 2025-11-12 17:50 UTC

10/11 Nov 2025, Rovaniemi. Parhelia sweeping by on the road that goes past the ski center. I should have tilted the phone higher up, though.

[video: 1988665680728846595-lw-fjSO8x6XvPK0s.mp4]

4/24 - 2025-11-13 13:12 UTC

10/11 Nov 2025. As I am about to embark on working the stacks, for reference I leave this account of the night's turns here. This is how I perceived it, exact details may get corrected/refined when I process the material. Let's see if any of those new halos I made an attempt at will turn up. The image shown is the first one I took with Canon.

* * * * *

Leaving the sledge behind saved a good part of the night. Even if conditions were still far too dry, I left apartment around 7pm. I thought killing time by continuing with the frozen puddles that I had taken a look at in the morning at the Ylikylä snow dump. Regarding the halos, it wasn't good, the plates were too thick and frosty for transmitted light. But I decided to try microscoping some pieces. Found Parry crystal growths here and there among mostly plate crystal growth. Some Parry groups were different, though, having not uniform width. I think they were frost crystals, sticking originally straight down from the ice plate, but fallen flat by the disturbance of breaking the piece. I have seen and photographed such fake Parry crystals last winter in my home grown plates.

Having the puddle crystals photographed, I left for the ski center, where on the highest bridge I assumed the waiting mode, walking back and forth. When after some hours at midnight inversion was still not developing (which you can judge from the temp difference between the airport and railway station) and it got a bit windy, I concluded that conditions are still some hours away and I might as well go back to apartment to have some sleep. Wake up at 2:30 and come back.

Right after having parked at my apartment house I realized that the sledge for hauling the equipment was not with me in the backseat. I had forgotten it at the snow dump. So back I drive, finding, to some relief, that the sledge had not yet found a new owner (there is some traffic in this location with people driving in and letting their dogs roam about). Another thing I find, to some surprise, is that the ski center is shrouded in ice fog.

Sleeping no more on agenda, I speed towards the ski center and arrive to a scene of beautiful Y-pillars. But once I have the camera up and running on the bridge, ice fog is already on the waning trajectory and soon completely gone. Back to the waiting mode, then.

But not for long. Maybe in half and hour or so I see pillars again rising up from the ski center lights, the ice fog soon engulfing the bridge where I have all set to capture the display right from the start. It came to stay, allowing to photograph until 6:00, which is when more lights are switched on for the cross-country skiers, involving also the one light on the bridge itself.

Without the lucky sledge accident, I would have missed a significant chunk of the action. But the core problem is of course the idea of going to sleep in the first place. If the forecast is for the conditions to get better, as it was, you stay, simple as that. Particularly more so as I was "fresh", not having like chased the three previous nights and days. Getting soft and puffy, then, weak for the siren calls of cookies and warm milk at home, the silky rustle of night pajamas.

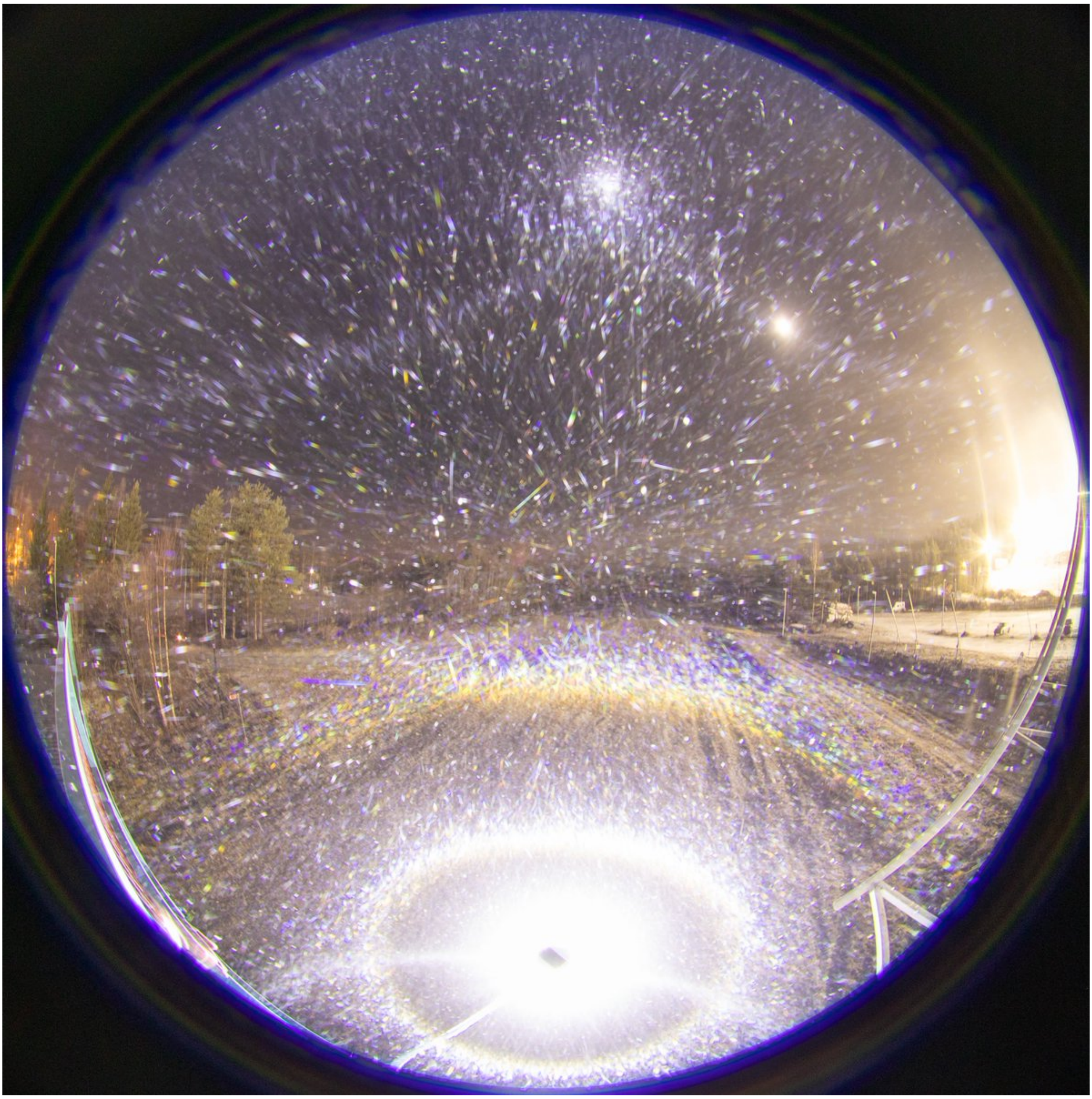
So, after having been forced out from the bridge, I took some videos of streetlight parhelia on the road passing the ski center. I also snapped some shots under one long-poled yellow streetlight next to Thulia restaurant at the ski center – passing cars are of no concern at that end-of-the-road spot. The 3D-view was awesome. Huge parhelic circle looming above, above and below it zenith and horizon arcs becoming in turn eye-detached and eye-connected as I adjusted my distance to the light. Possibly the best one I have seen. Unfortunately, when I started taking photos, it was waning already (haven't we heard this before). And with the dawn already cracking at the horizon, there was no reason for waiting it to get back. Of course the 3D character is lost in photos vs video, but still, they have their worth.

At the beginning, the ski center red digital display was undecided between -8 and -9 C, then more stable -9 C, at six o'clock I noticed -10 C reading. Initially, it seemed like column only, possibly with minimal plate contribution. Some Parry must have been involved because I could distinguish a weak X in the ski center lights. Then at some point plate was definitely added to the mix, Moilanen arc also entering the company, the column, in turn, taking a lesser role. When the lights got on at 6, the swarm was heavily plate dominated, which is perfectly in line with the falling temp.

My plan was to capture the still theoretical flip upper horizon arc (plate version 312 , Parry 136, triangular Parry 14846). Another halo I tried to capture was upper deepcave Parry (raypath 64), the last remaining undiscovered member of the Parry arc sextet. I was not able to spot either of these visually, or in the cameras. The light pollution is certainly a problem for weak features. But let's see about the stacks.

Another thing in my mind was the extreme elevation arc that has reverse raypath (148) to the Ounasvaara arc. Given how minor the Parry orientation contribution seemed on the basis of the weak helic arc on the ski center lights, I don't think there is chance for its discovery (moreover, it needs triangular shapes, which may have not been present).

I also started taking elevation series of the Moilanen arc when I realized it had entered the play. We don't have photos of mid and extreme negative light elevation Moilanen arcs. Down from the horizon, its visibility caps around -72 degs light elevation, thus having a more extensive range down than upwards from horizon. Who knows if Moilanen crystals can produce some unforeseen halos at this unexplored negative range land.



5/24 - 2025-11-13 17:19 UTC

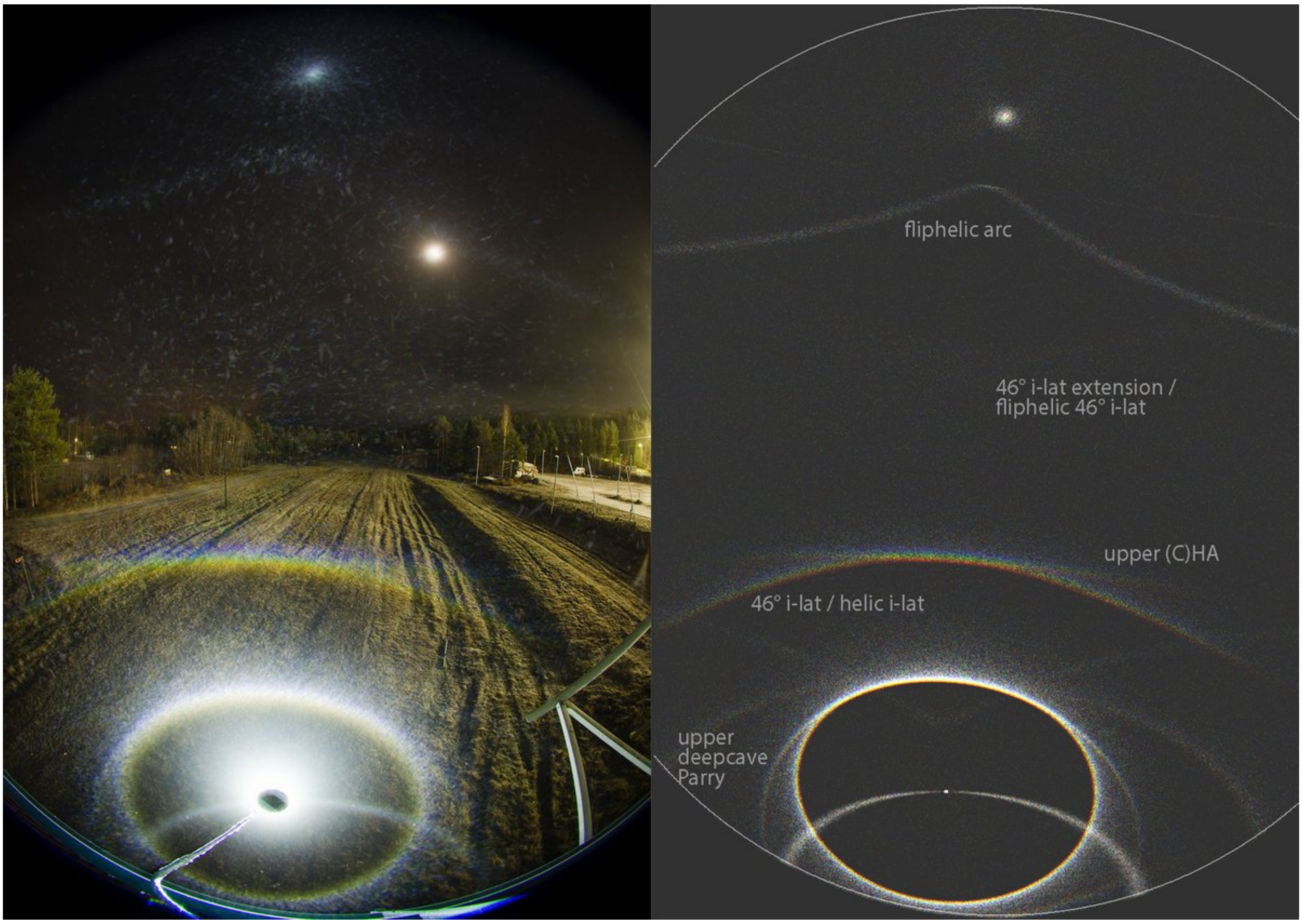
The raypath 64 I gave for the upper deepcave Parry is wrong. It is 53. Raypath 64 is the lower shallowcave Parry, which has been photographed. Of course these are also made by the vertically mirrored raypaths 68 and 73.

6/24 - 2025-11-14 07:53 UTC

10/11 Nov 2025, Rovaniemi. This is the first publishable set. The upper deepcave Parry (53) is there, completing the Parry sextet. I don't quite see the flip upper horizon arc here, only the 46° infralatarc extension / fliphelic 46° infralatarc. The simulation is for -67, but there is room for error here. Parameters in the next post.

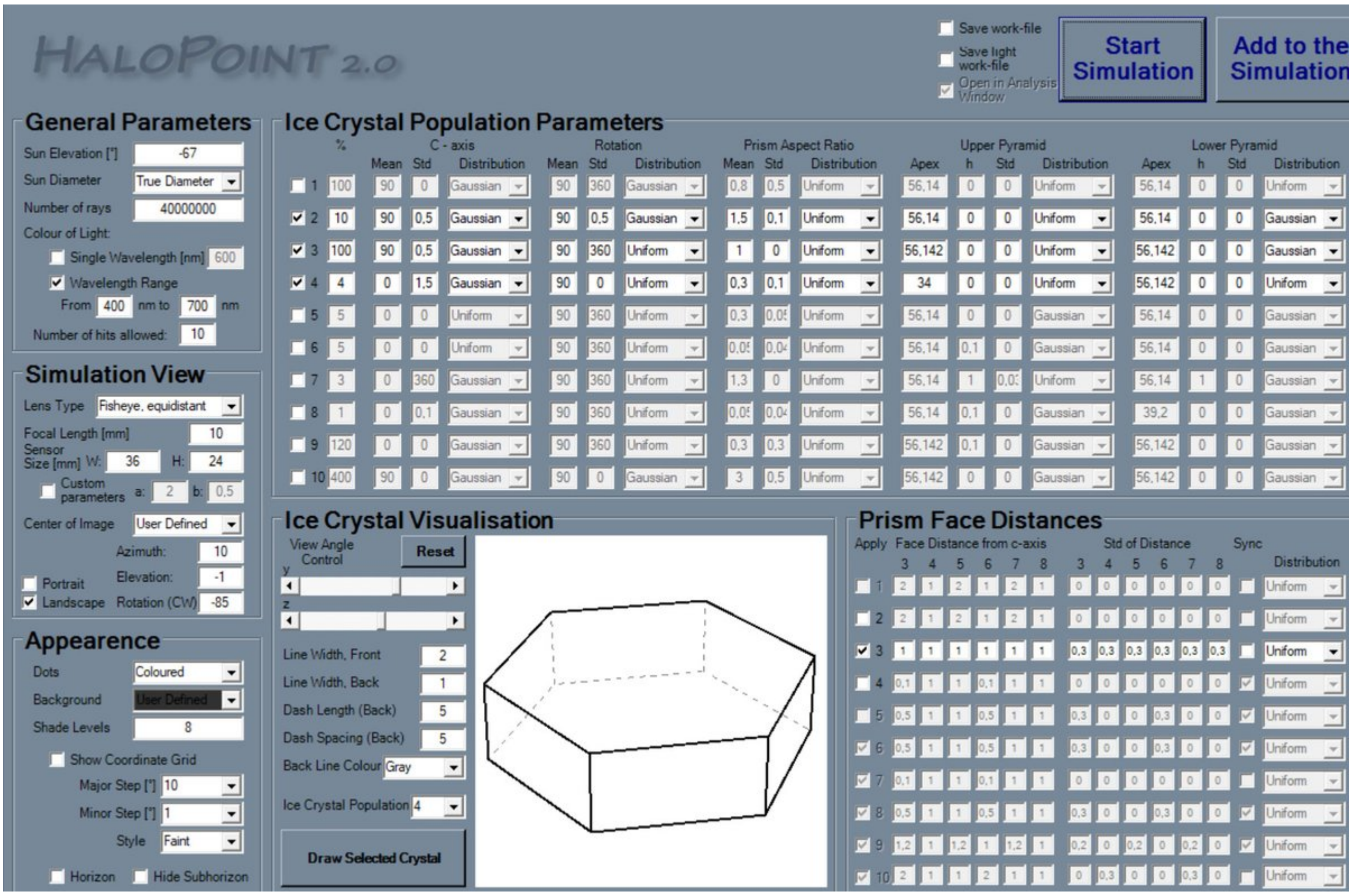
The beam of Olight Javelot Turbo was a bit front-loaded, making, among other things, the tanarc weaken towards nadir.

22f, 20s, 1:47-1:56 am. Nothing to publish from the first wave because the display was fizzling out when I got the photography going. Between the first and second wave there was 40 minutes break.



7/24 - 2025-11-14 07:55 UTC

10/11 Nov 2025, Rovaniemi. Parameters for the previous post's simulation.



8/24 - 2025-11-15 11:44 UTC

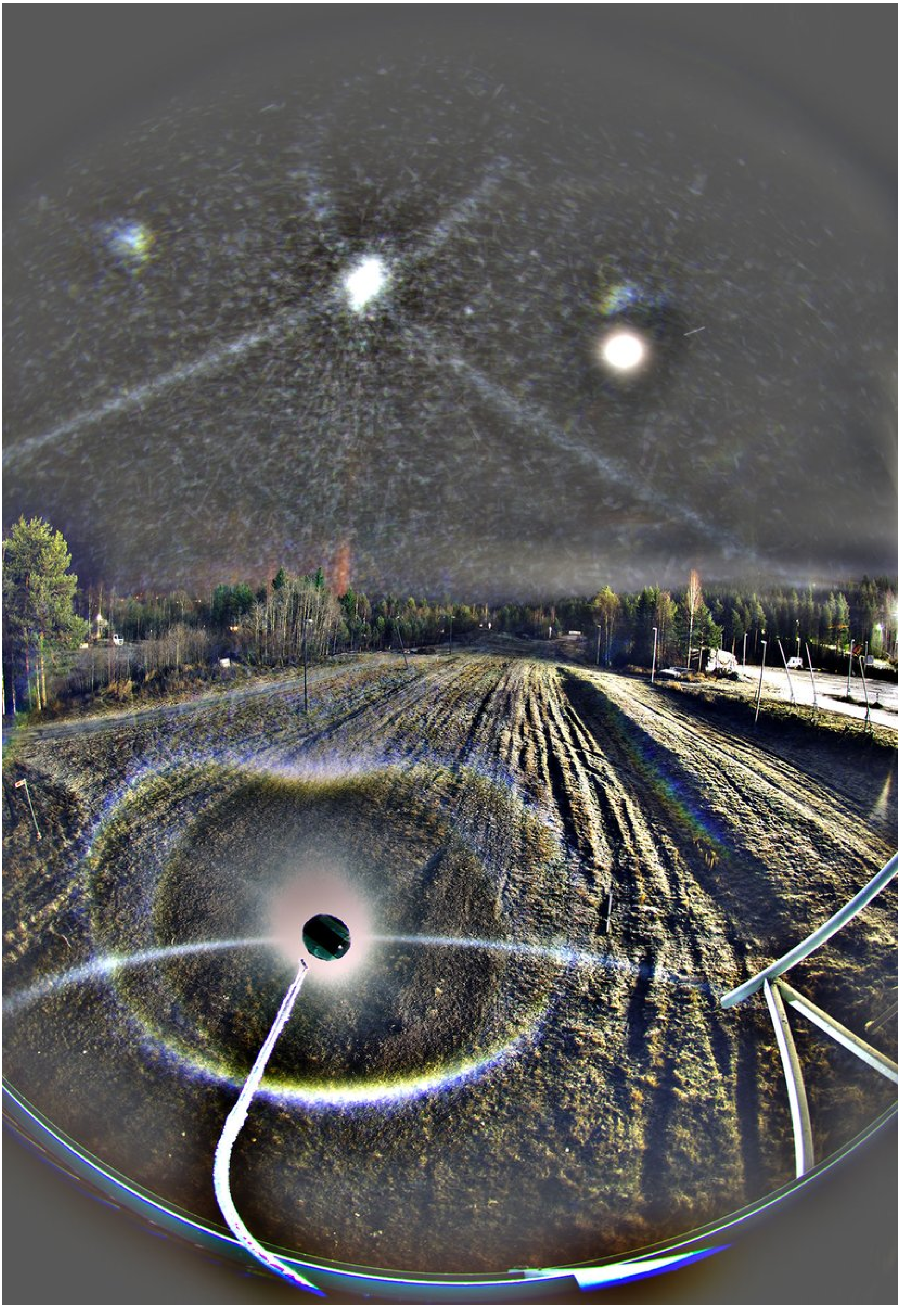
10/11 Nov 2025, Rovaniemi. 34 x 20s at 02:02-02:14.

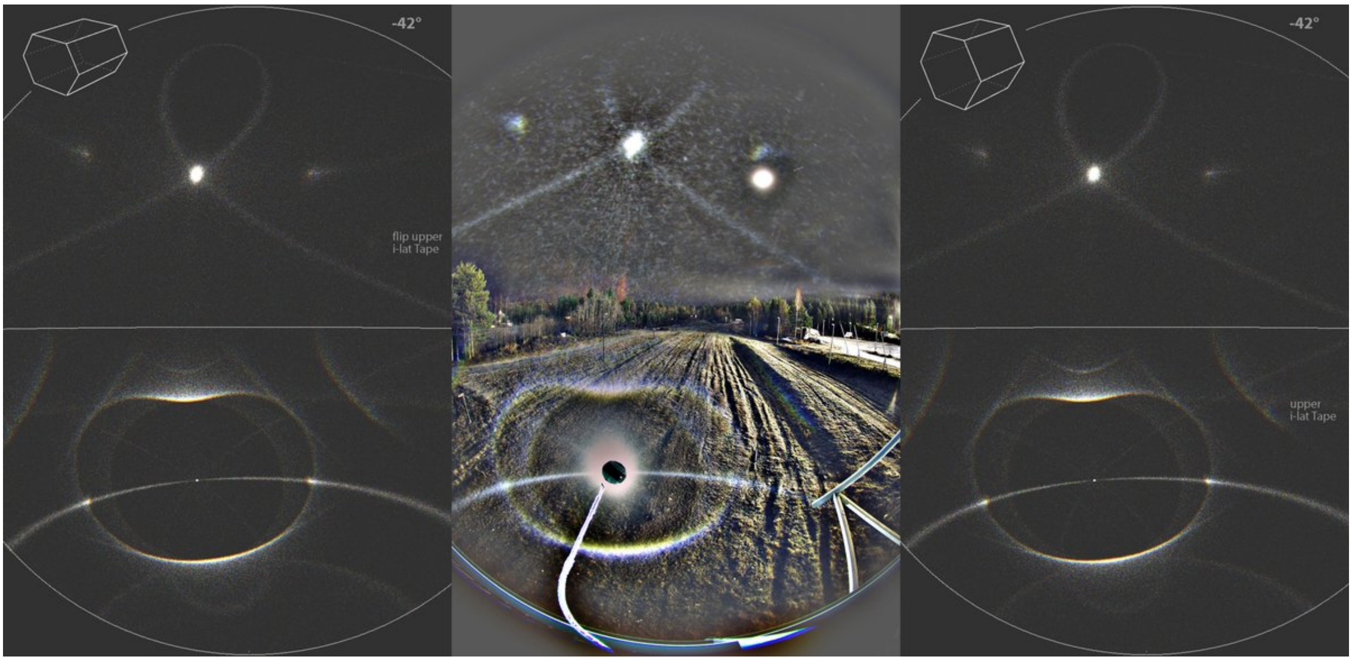
The next lamp-camera configuration I had was for around -42°, as estimated by the location of parhelia slightly inside circumscribed halo. One detail catches the eye: at five o'clock from the moon is a halo-like spot. As it happens, simulations produce in this location Parry 135/237 – "flip upper infralat Tape", if you will. Could that really be? It looks real, made by separate crystals, as shown by the max stack and close-up of the average stack in the next post.

Too bad we have only one side to work with. When in the Rovaniemi display on the night of 6/7 January 2016 we found, in an unexpected location, a halo-looking feature whose instance on the other side could not be seen due to asymmetric view, and I started blabbering excitedly about a new halo, Lefa – who was on the 4-man team chasing that night – cut such nonsense short by asking "What new halo?" His point was that you need to have it on both sides before an artefact can be discounted. But as it turned out, such qualms could be put to rest a couple of days later when processing of another set of photos taken three hours later showed the same anomalous arc – and this time the view was symmetric with both the left and right instance of the arc visible.

Anyway, as to the display at hand, the Lefa's "both sides rule" perhaps does apply so strongly due to the arc visibly being made of individual crystals and the possibility to compare the location to a simulation. Neither of which applied to the 2016 display arc.

If this indeed is a true flip upper infralat Tape, then the crystals need to have been tabular. Even if the evidence in halo displays for tabulars is not as solidly established as for the triangular shape continuum, now and then we do need to summon tabulars in halo simulations to explain some features. And here tabulars clearly produce better simulation than regular hexagons, as shown by the two simulations of the last image.



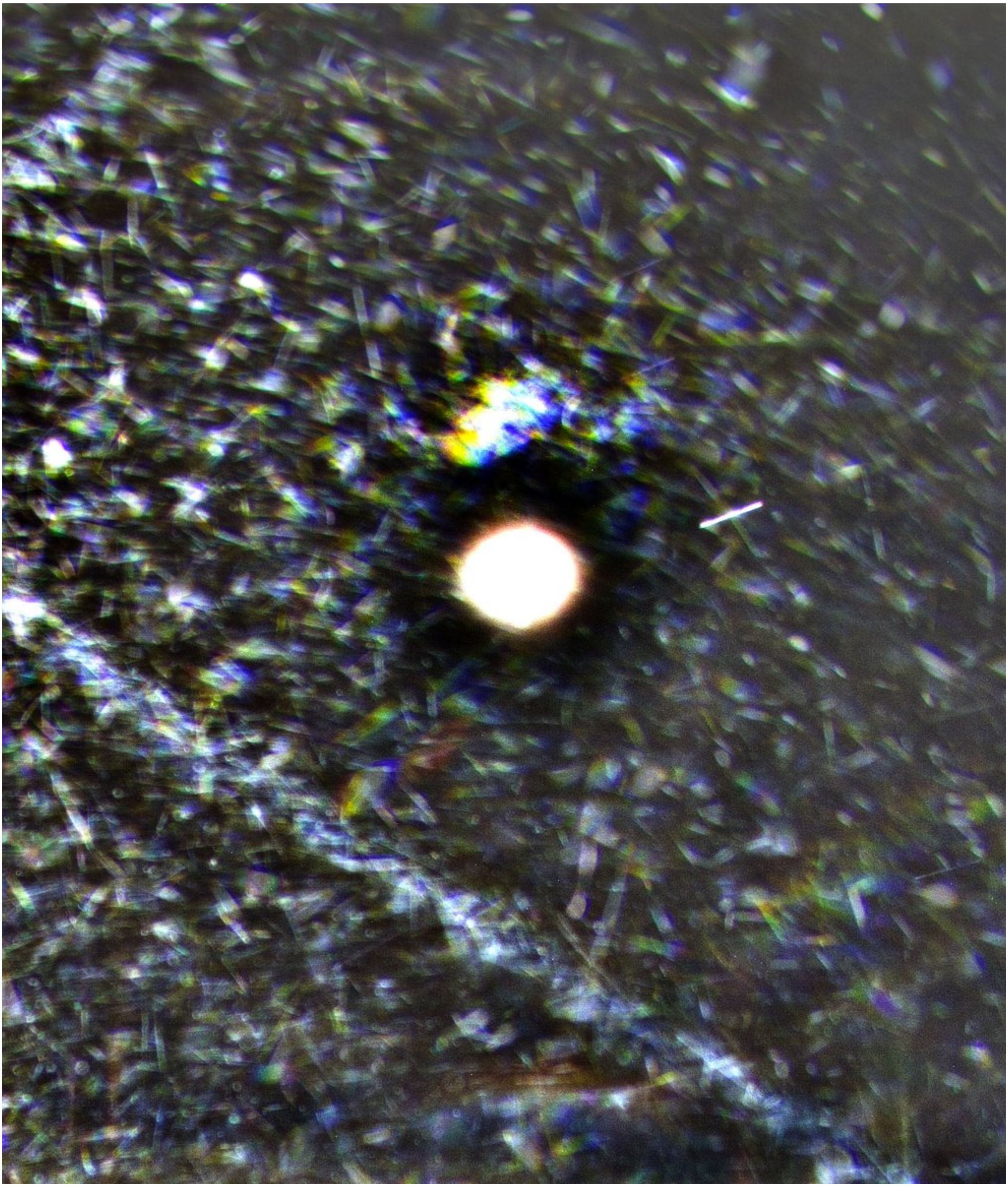


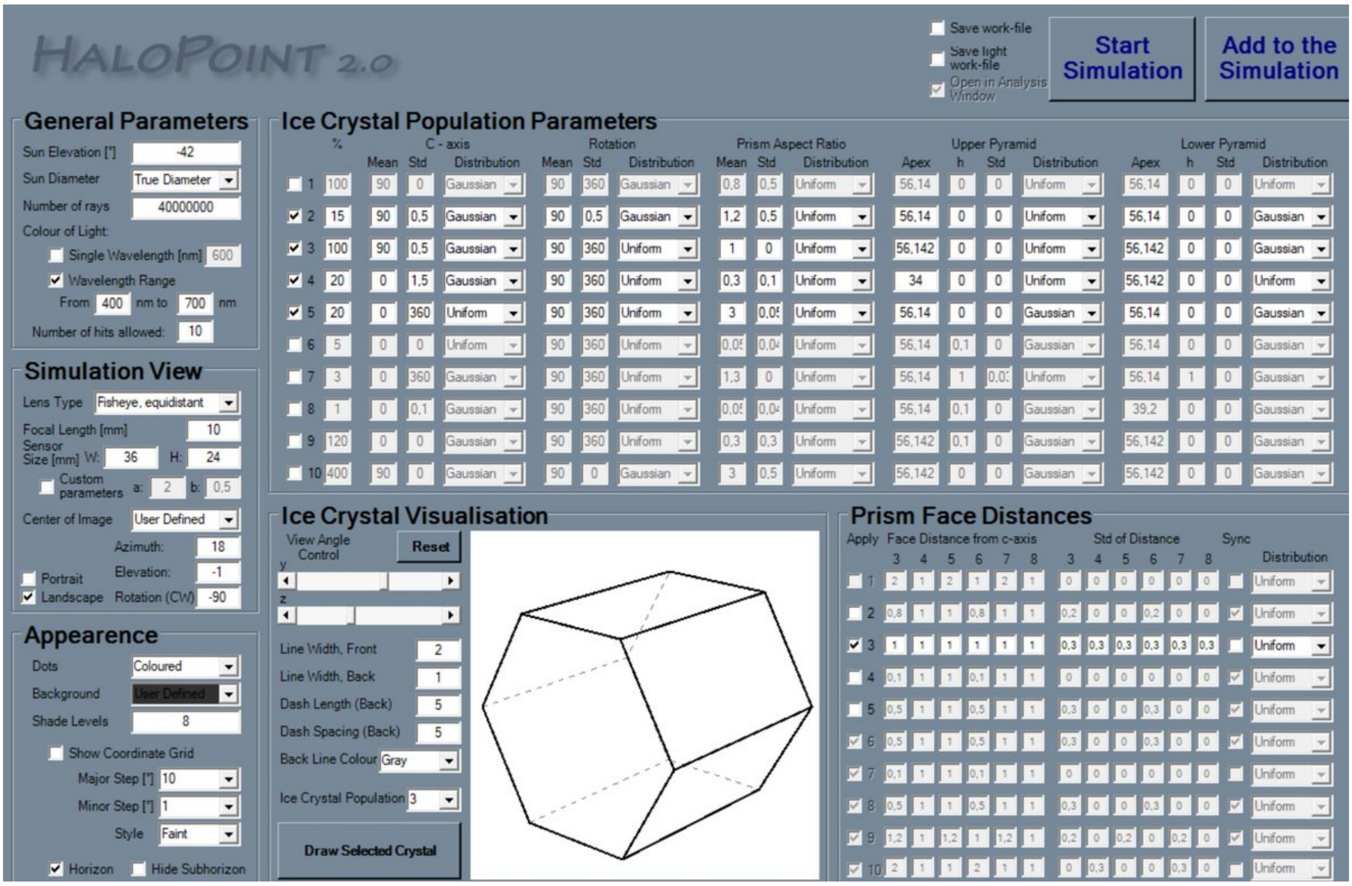
9/24 - 2025-11-15 11:46 UTC

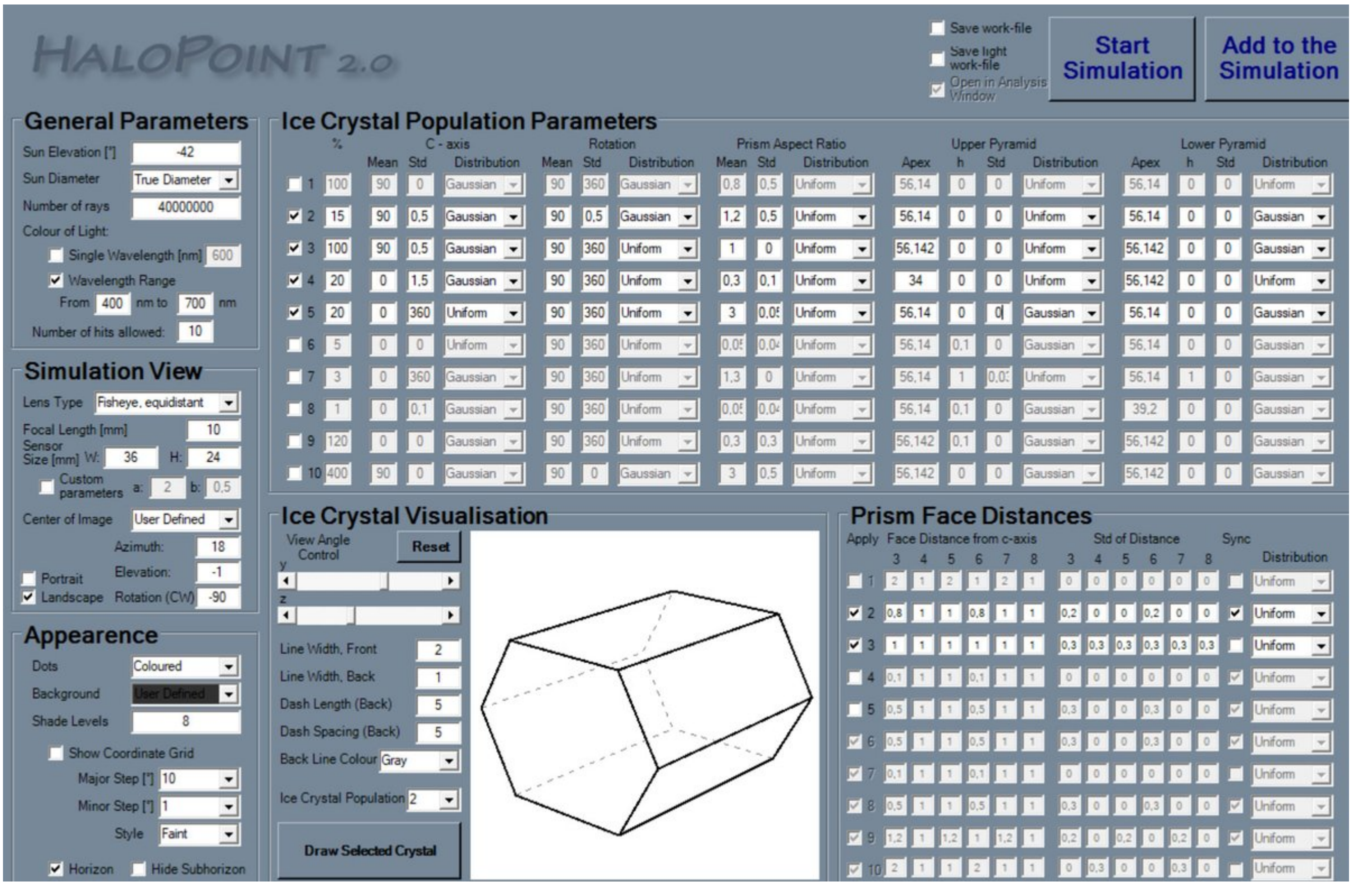
10/11 Nov 2025. The first image is a close-up of the feature in ave stack, the second image a max stack of all 34 frames – the signal is present in this stack, too. The other two images are parameters for the two simulations in the previous post.

While a horizontally lamp centered view would have been optimal, I wonder why I didn't choose the second best option, lop-siding towards the darker direction? Why I had to choose the worst of the three options, lop-siding towards the light pollution? What I have run in FastStone Image Viewer quickly though all my Nikon photos from this night, the view in all of them seem to more or less biased towards the slope lights as if by magic attracting force. This is even more worrying because I was aware at the time that the view was wrong. Chalk it up on the brain fog of the season's opening night.

Also chalk it up on the brain fog that I was thinking only of capturing the upper deepcave Parry, flip (circum)horizon arc, and the "reverse Ounavaara arc". The prospects for flip Tape and Parry arcs didn't cross my mind. It is not like I have not been aware of their possibility in the bridge set-up. Many a times I have thought about their capturing.

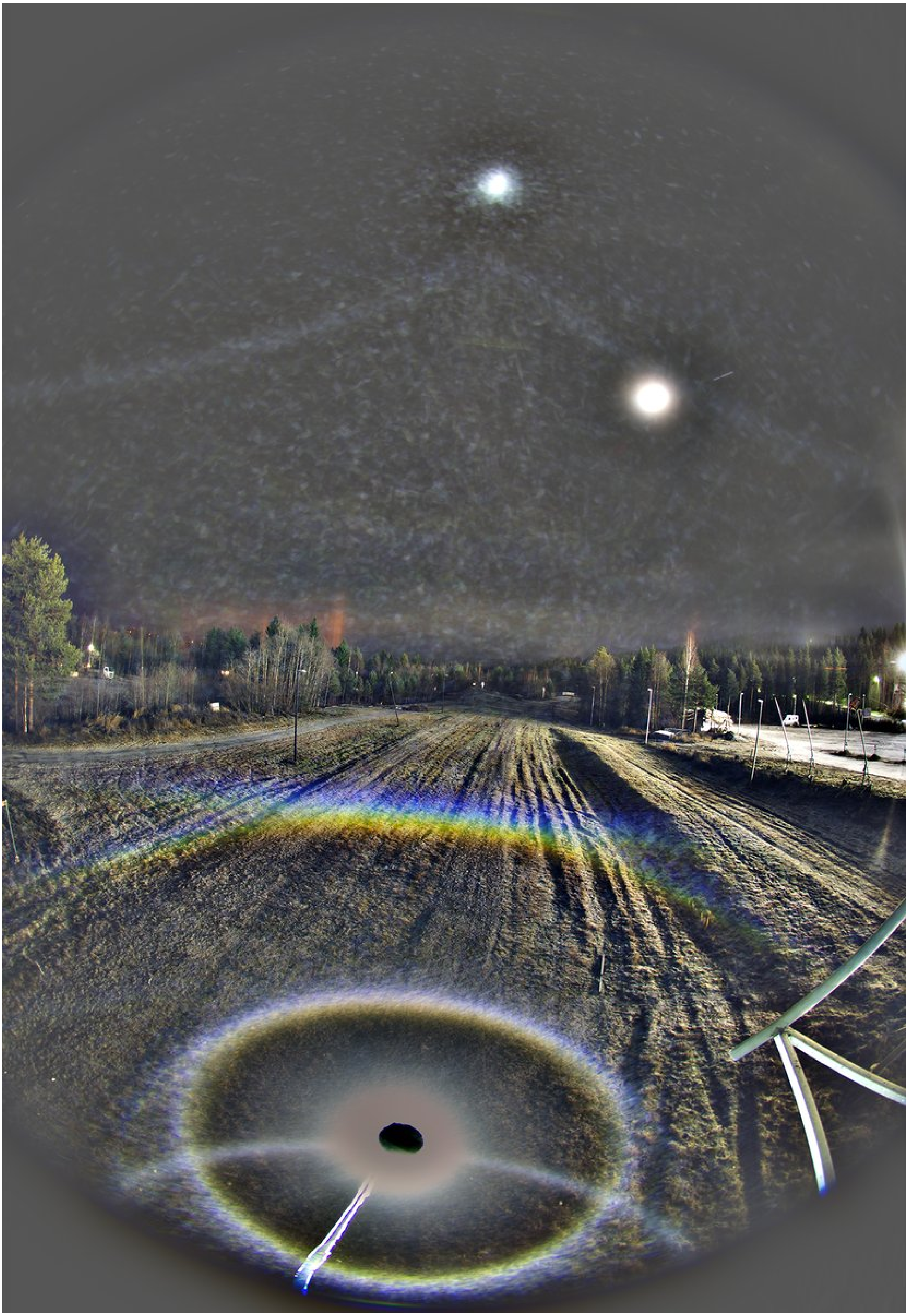


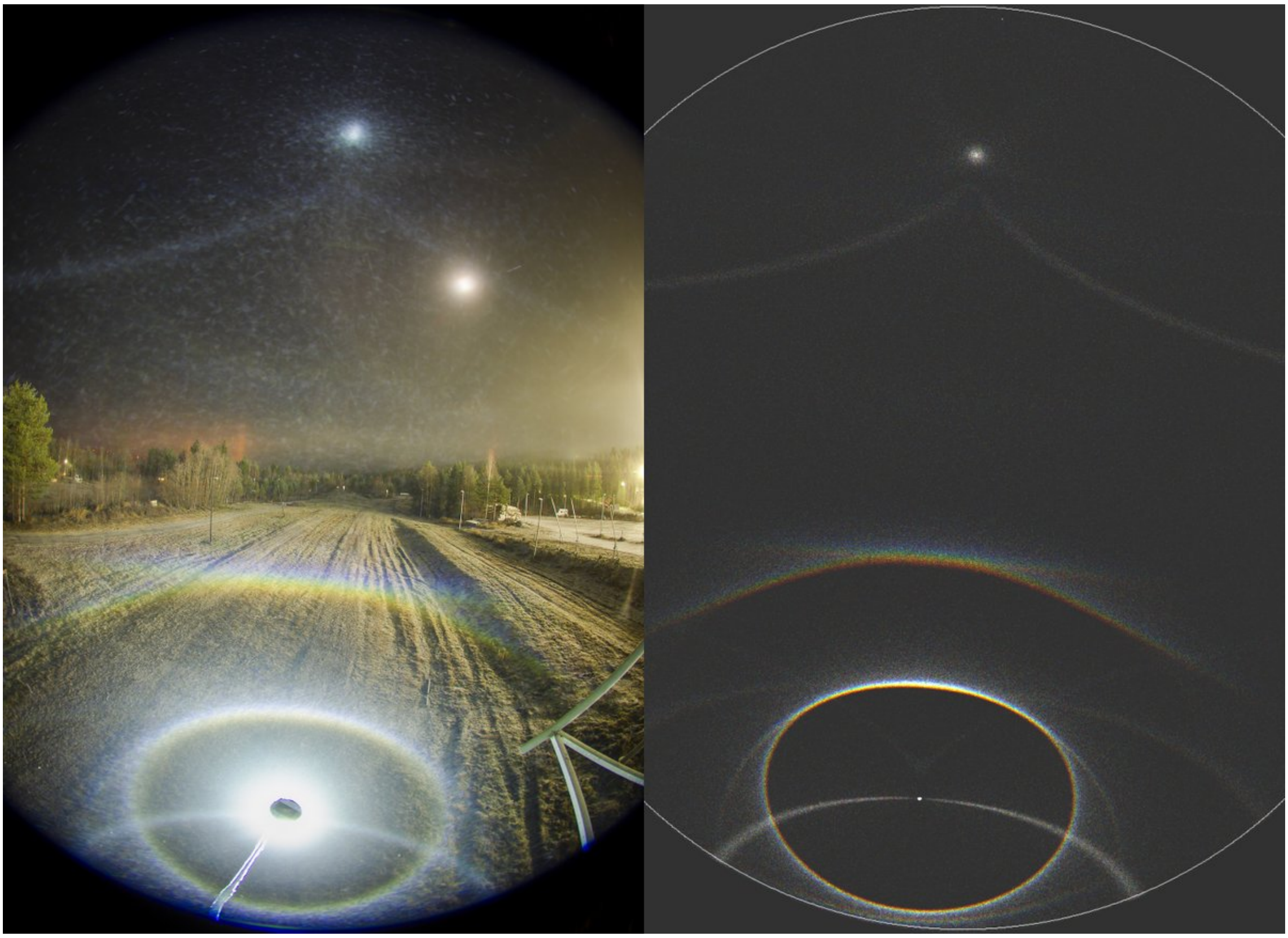

HaloPoint 2.0
Save work-file
Save light work-file
Open in Analysis Window
Start Simulation
Add to the Simulation
General Parameters
Sun Elevation [°] -42
Sun Diameter True Diameter
Number of rays 40000000
Colour of Light:
Single Wavelength [nm] 600
Wavelength Range
From 400 nm to 700 nm
Number of hits allowed: 10
Simulation View
Lens Type Fisheye, equidistant
Focal Length [mm] 10
Sensor Size [mm] W: 36 H: 24
Custom parameters a: 2 b: 0,5
Center of Image User Defined
Azimuth: 18
Elevation: -1
Portrait
Landscape Rotation (CW) -90
Appearence
Dots Coloured
Background User Defined
Shade Levels 8
Show Coordinate Grid
Major Step [°] 10
Minor Step [°] 1
Style Faint
Horizon
Hide Subhorizon
Ice Crystal Population Parameters
% C - axis Rotation Prism Aspect Ratio Upper Pyramid Lower Pyramid
Mean Std Distribution Apex h Std
Ice Crystal Visualisation
View Angle Control
Reset
Line Width, Front 2
Line Width, Back 1
Dash Length (Back) 5
Dash Spacing (Back) 5
Back Line Colour Gray
Ice Crystal Population 3
Draw Selected Crystal
Prism Face Distances
Apply Face Distance from c-axis Std of Distance Sync Distribution


HaloPoint 2.0
Save work-file
Save light work-file
Open in Analysis Window
Start Simulation
Add to the Simulation
General Parameters
Sun Elevation [°] -42
Sun Diameter True Diameter
Number of rays 40000000
Colour of Light:
Single Wavelength [nm] 600
Wavelength Range
From 400 nm to 700 nm
Number of hits allowed: 10
Simulation View
Lens Type Fisheye, equidistant
Focal Length [mm] 10
Sensor Size [mm] W: 36 H: 24
Custom parameters a: 2 b: 0,5
Center of Image User Defined
Azimuth: 18
Elevation: -1
Portrait
Landscape Rotation (CW) -90
Appearence
Dots Coloured
Background User Defined
Shade Levels 8
Show Coordinate Grid
Major Step [°] 10
Minor Step [°] 1
Style Faint
Horizon
Hide Subhorizon
Ice Crystal Population Parameters
% C - axis Rotation Prism Aspect Ratio Upper Pyramid Lower Pyramid
Mean Std Distribution Apex h Std
Ice Crystal Visualisation
View Angle Control
Reset
Line Width, Front 2
Line Width, Back 1
Dash Length (Back) 5
Dash Spacing (Back) 5
Back Line Colour Gray
Ice Crystal Population 2
Draw Selected Crystal
Prism Face Distances
Apply Face Distance from c-axis Std of Distance Sync Distribution

10/24 - 2025-11-15 19:28 UTC

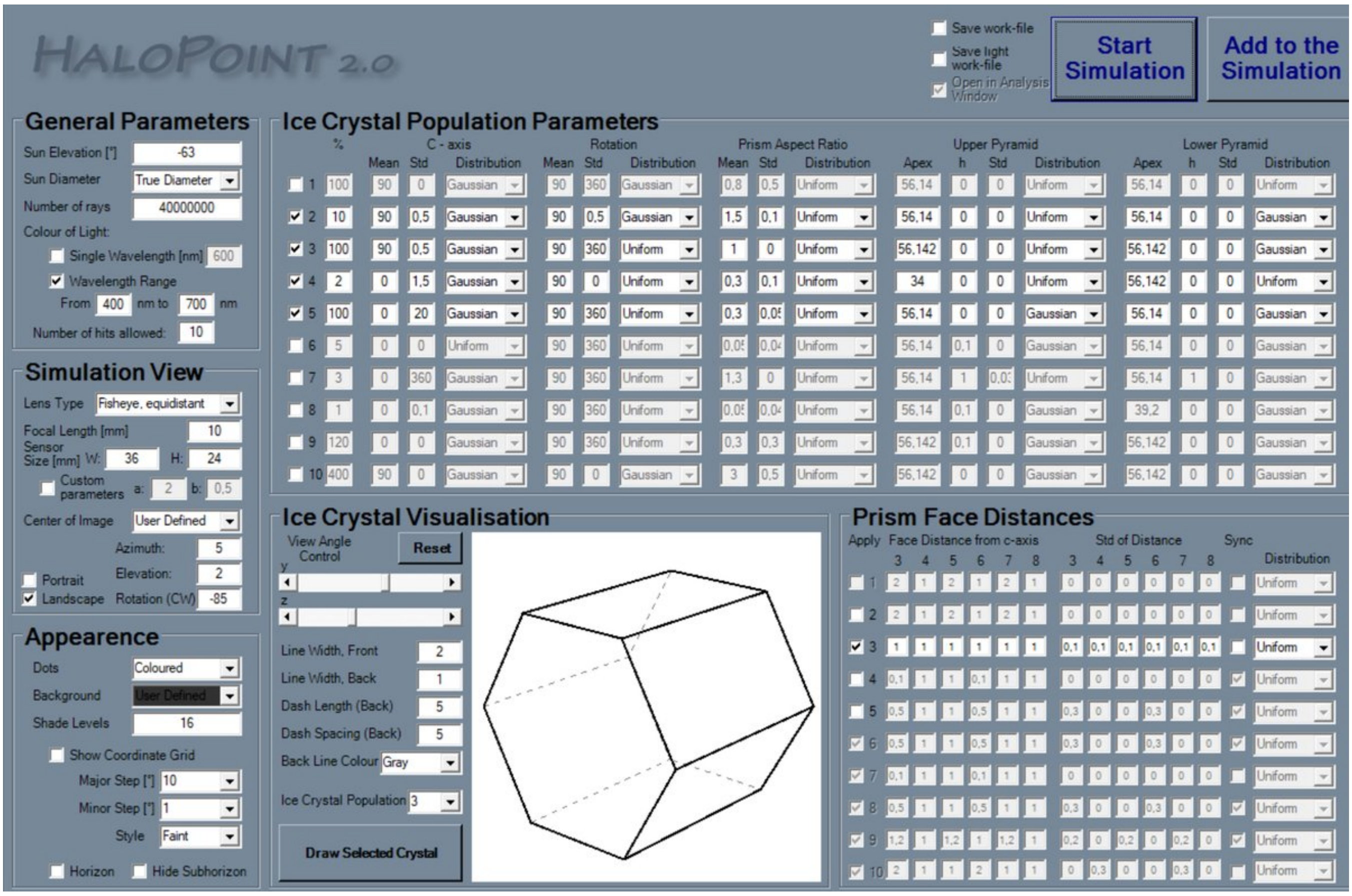
10/11 Nov 2025, Rovaniemi. 32 x 20s, 02:18-02:30. Nothing to report here, there may be a suggestion of flip upper horizon arc here, but can't quite claim it. At the upper right, just below the level of moon, is something that could be zenith arc from the slope lights. In the next post max stack and sim parameters. I just noticed the jpg was at the 60% quality, so I need at some point take a look if this has affected the previous images. This came probably because I changed to the newest version of the Photoshop.





11/24 - 2025-11-15 19:29 UTC

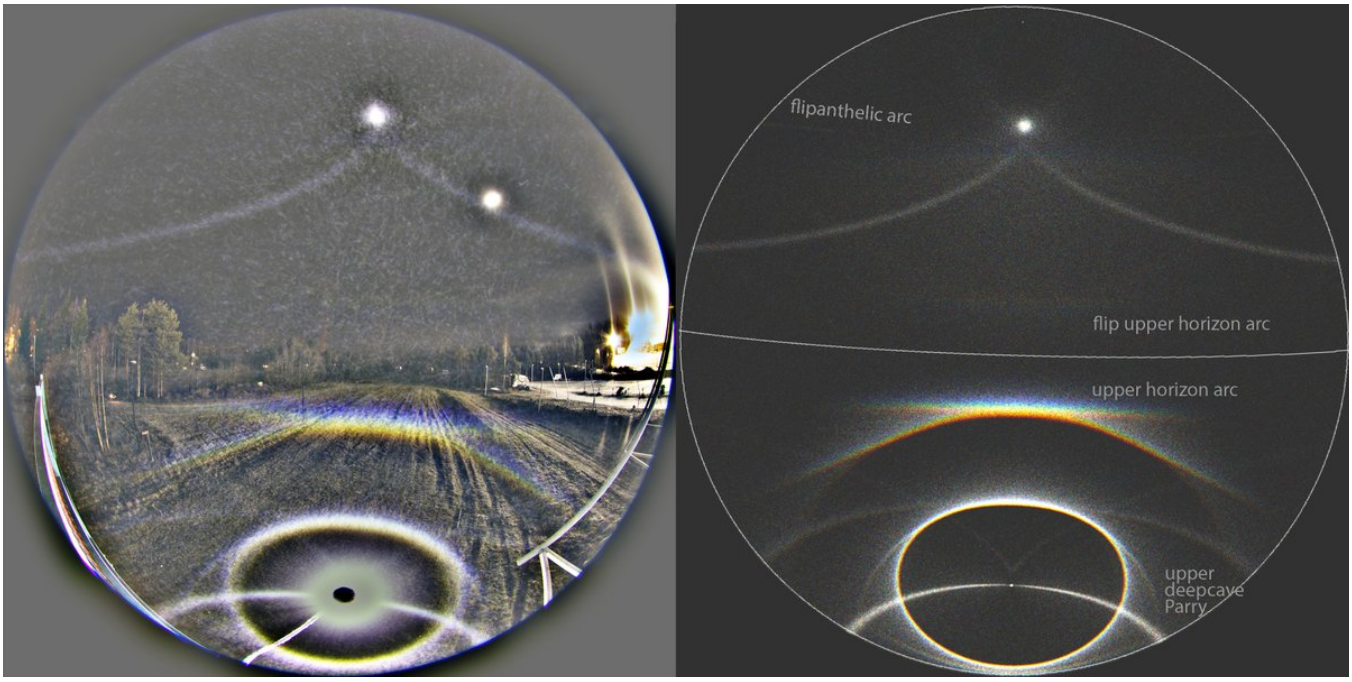
10/11 Nov 2025, Rovaniemi. 32 x 20s, 02:18-02:30. Blue circle came out real nice in this max stack. Parameters are for the previous post's sim.

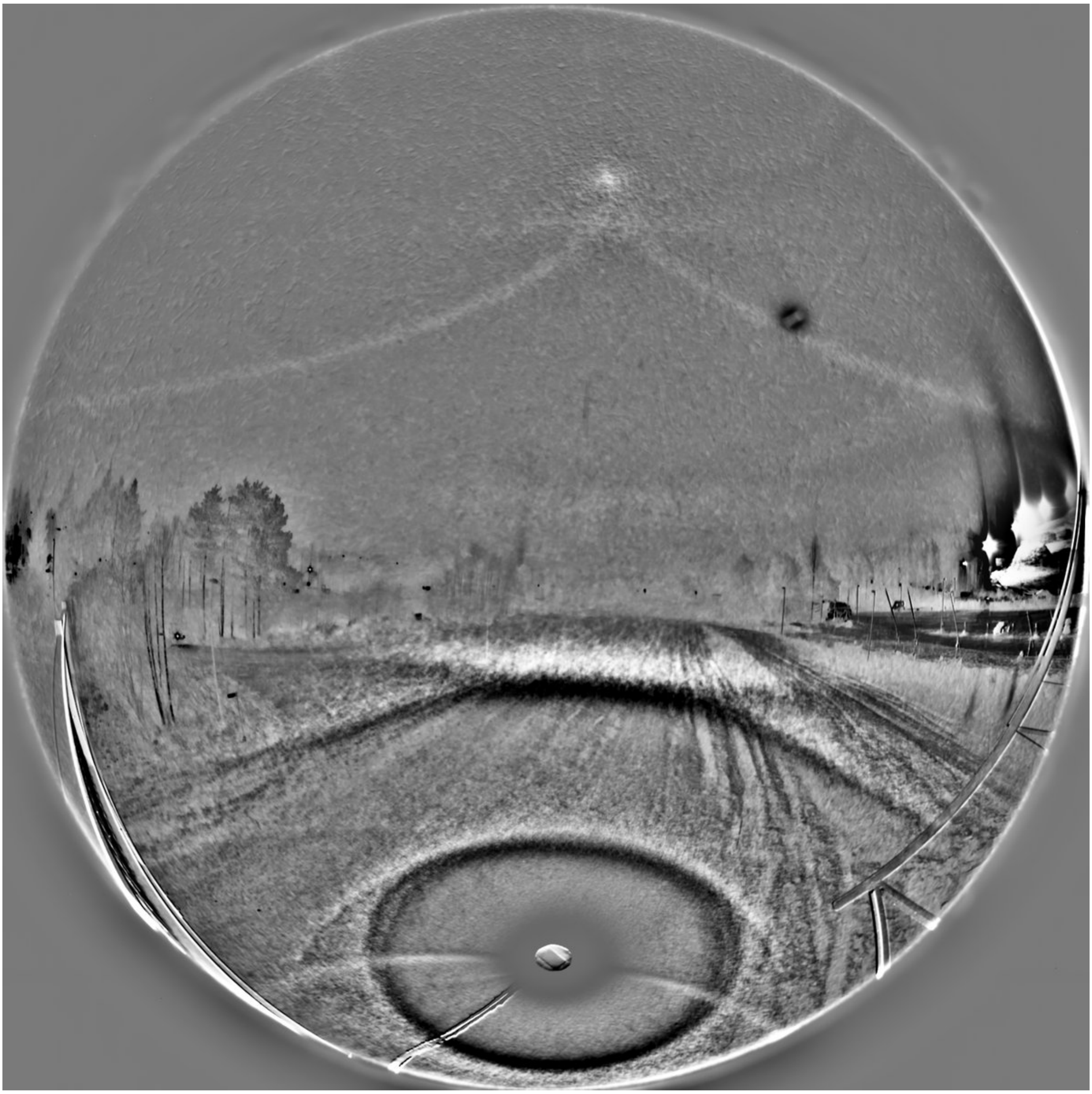


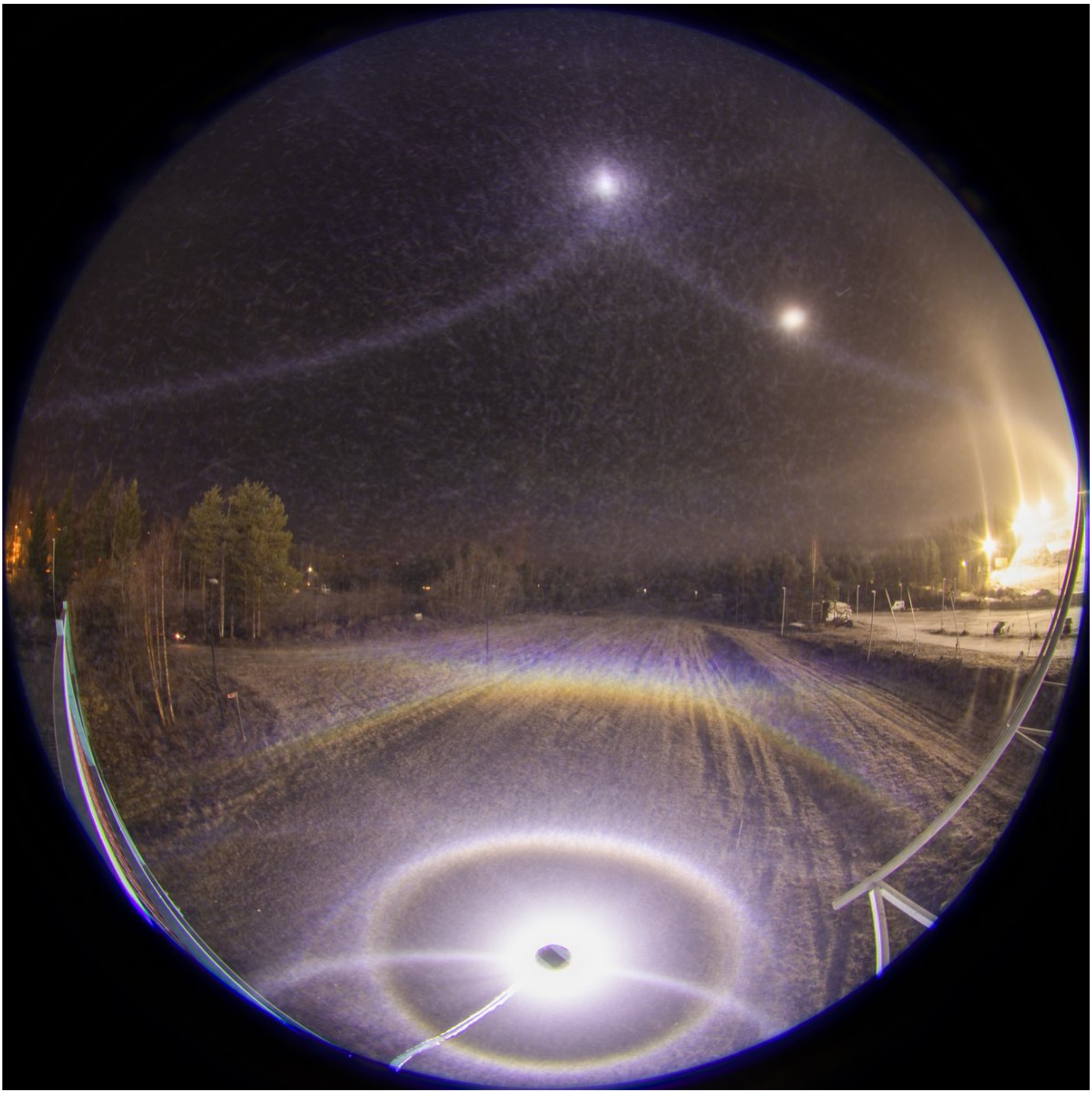
12/24 - 2025-11-15 21:31 UTC

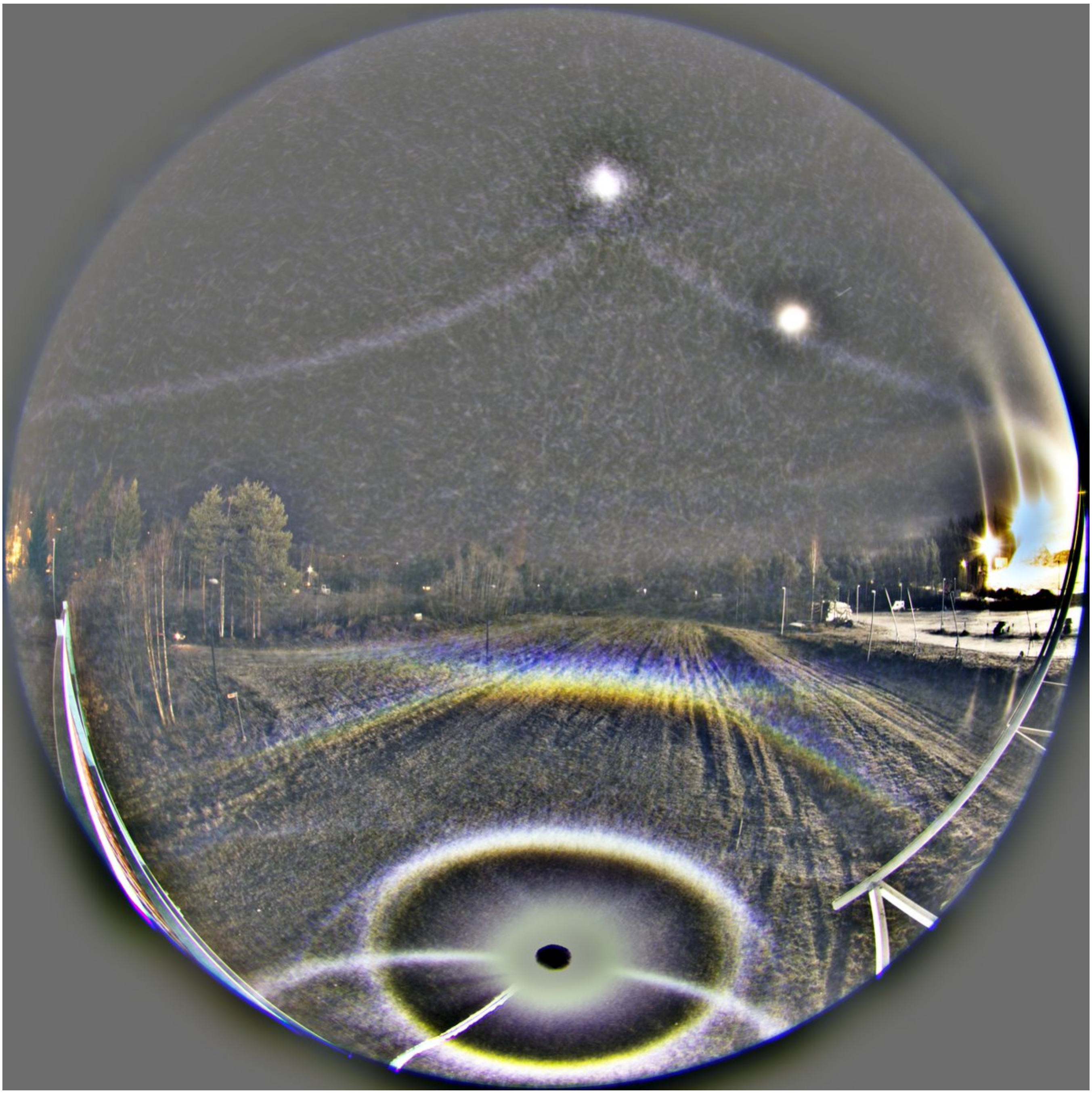
10/11 Nov 2025, Rovaniemi. 69 x 10s, 02:49-03:04. Continuing chronicling the night in chronological order. At this point I had taken Nikon in the idling car to warm up. Wisely done, you do it before the lens frosts – a line to which I managed to hew on this particular occasion.

Anyway, with this Canon 80D stack at the turn of the third hour I am emboldened to leave for the board of respect halo judges the claim for the flip upper horizon arc, or whatever name it should be called.


flipanthelic arc
flip upper horizon arc
upper horizon arc
upper
deepcave
Parry

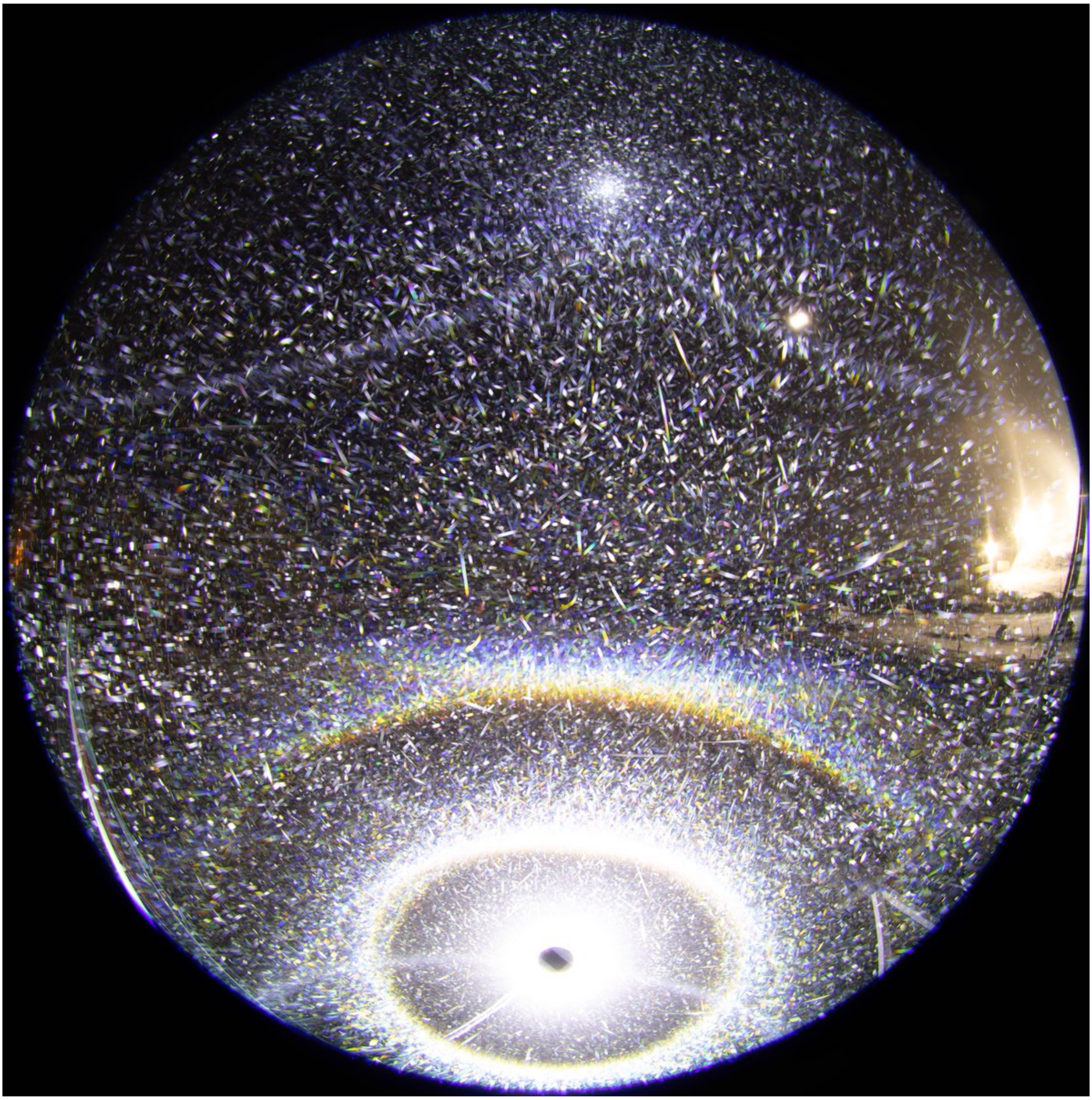


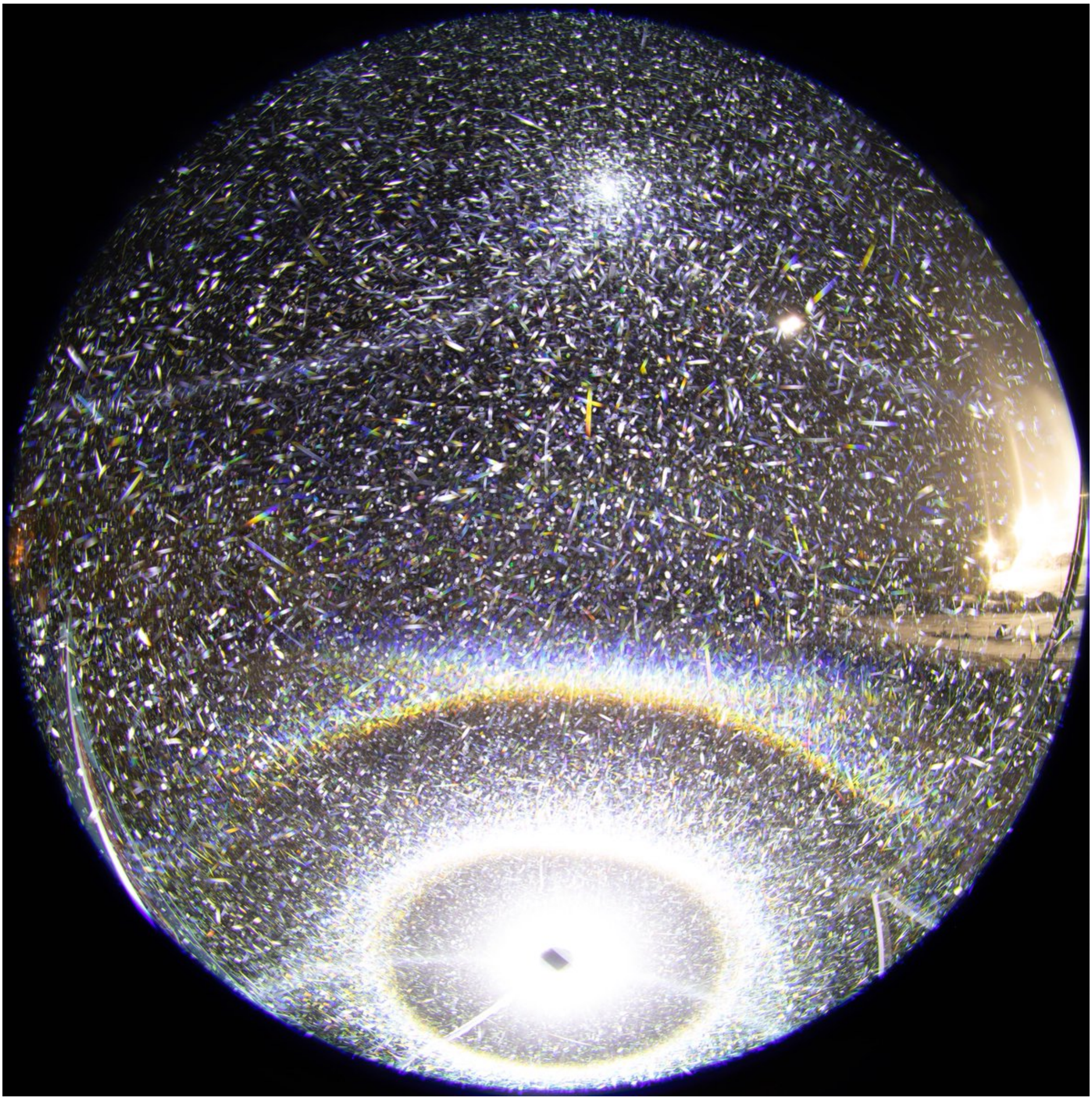


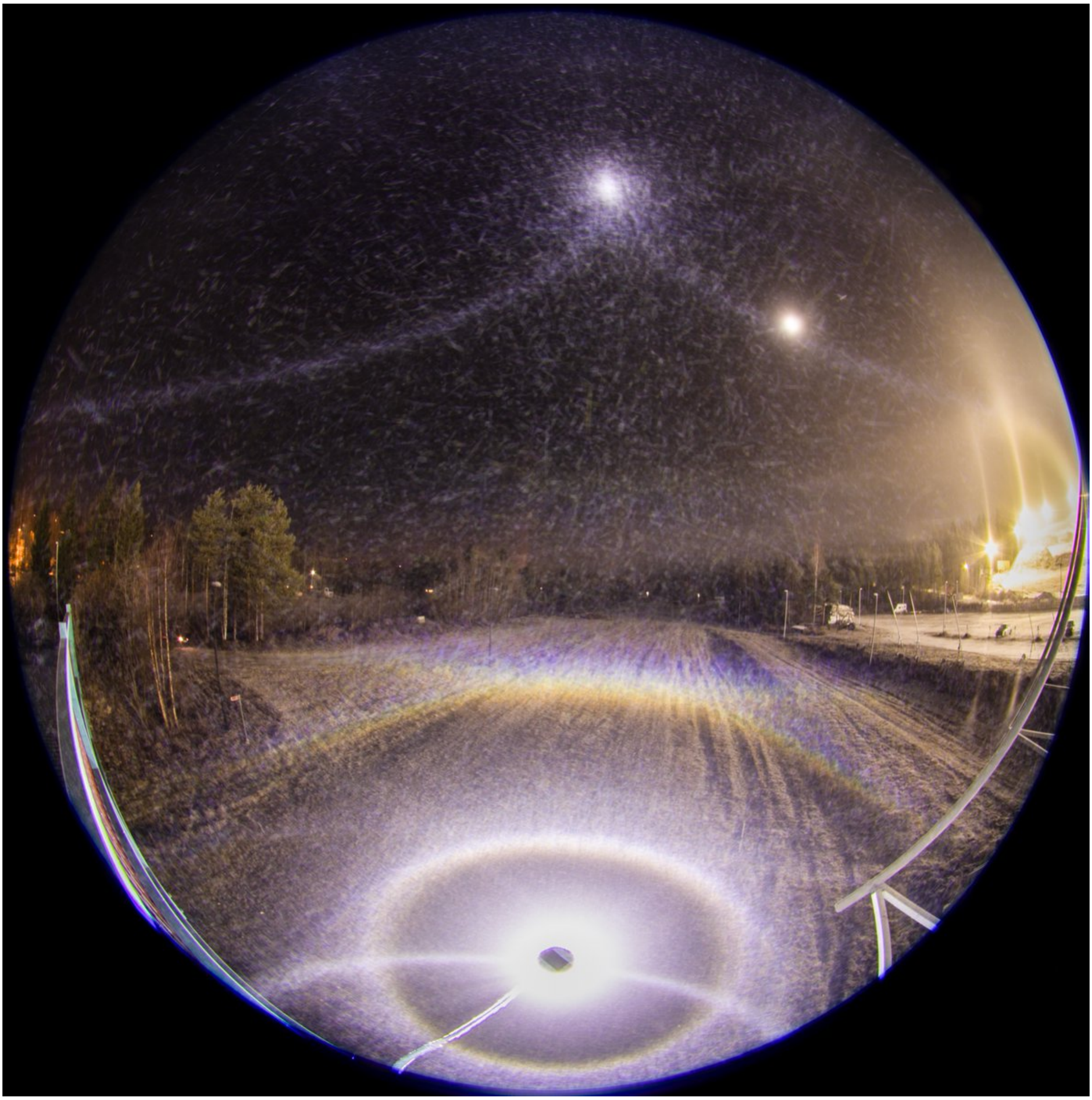


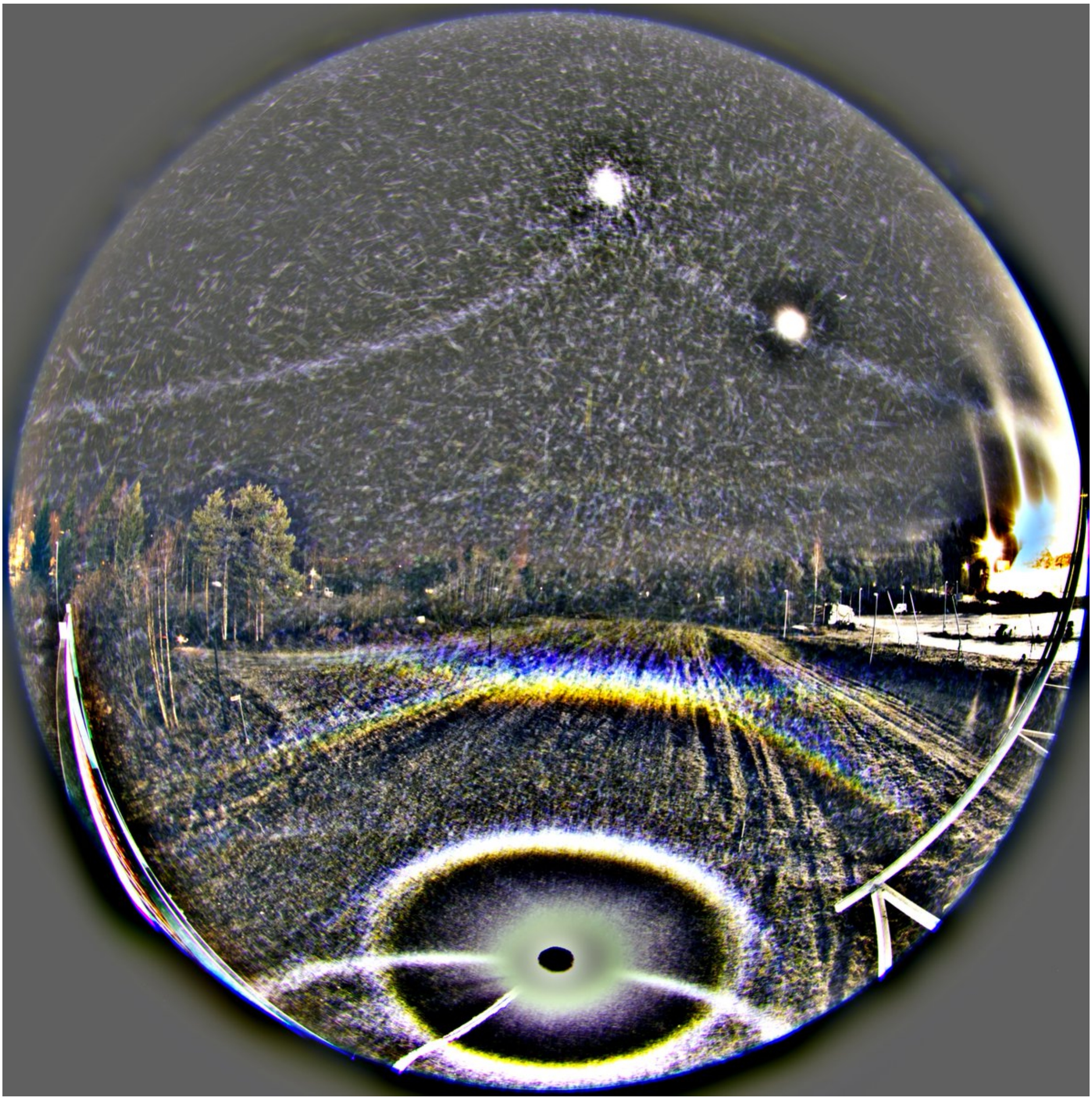
13/24 - 2025-11-26 08:29 UTC

10/11 Nov 2025, Rovaniemi. Max stack of the halves of the 69 set above. The flip upper horizon arc is visible in the first half but not quite so in the second. So I did an ave stack of the first half (two last images), but can't say this is really an improvement to the full stack. In the second max stack blue circle stands out better.





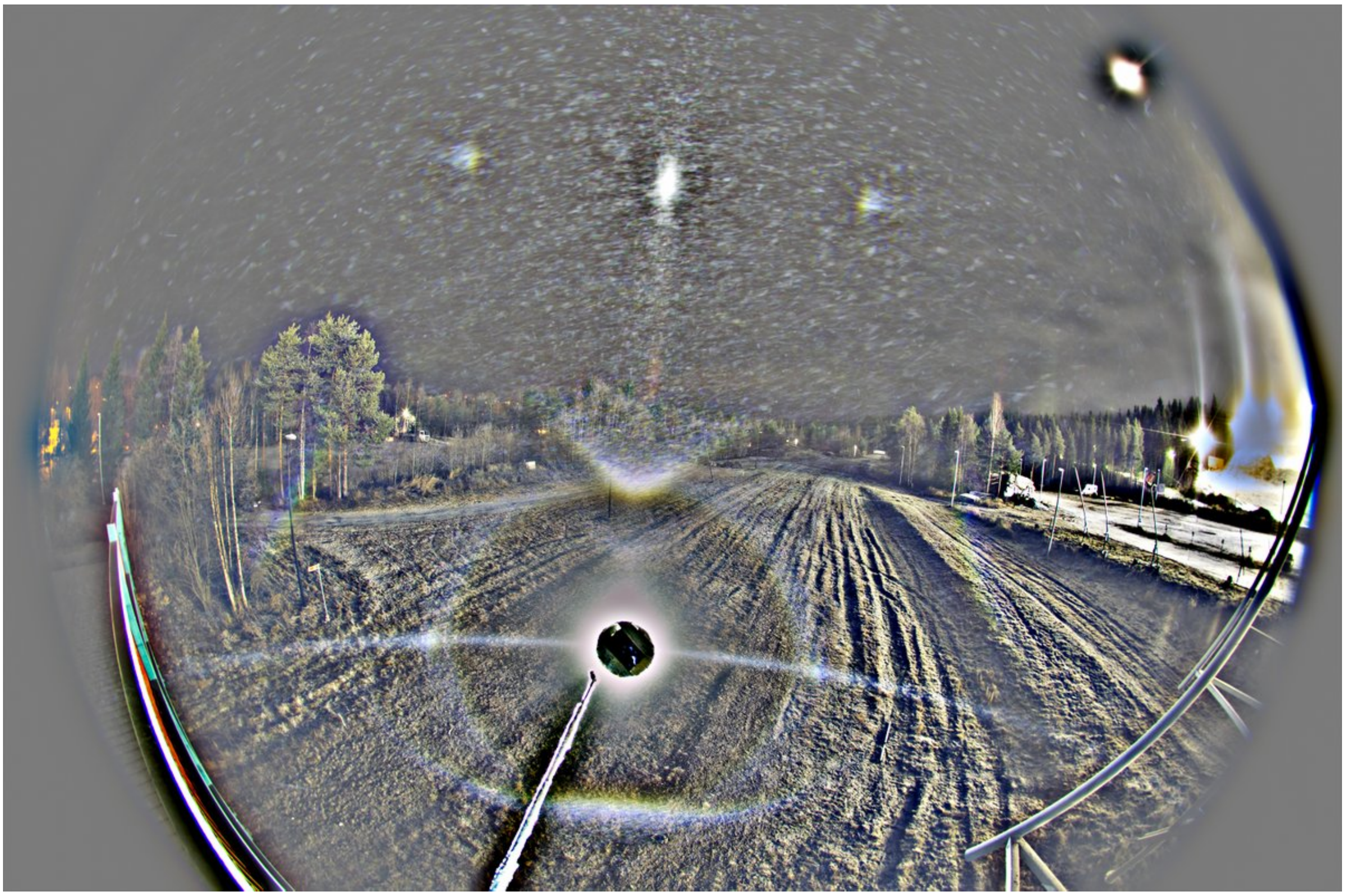




14/24 - 2025-12-04 08:52 UTC

10/11 Nov 2025, Rovaniemi. After 4 am, enter Moilanen. Is this the lowest elevation Moilanen arc on record? Light is thereabouts of -30°. Ski center lights are really disturbing, max is better.

The spotlight displays have increased the light elevation range from about 95 to full 180 degrees and that has put the vocabulary in crisis. What is, for example, mid-elevation? Is it both sides of the horizon at around 30-50°, or is it around 0°? And suppose you want to describe some arc's behaviour. What does it mean if you say "with increasing light elevation..." when you are talking about the negative range? Or if you want to cover the identical action on both sides? And so on and so on.

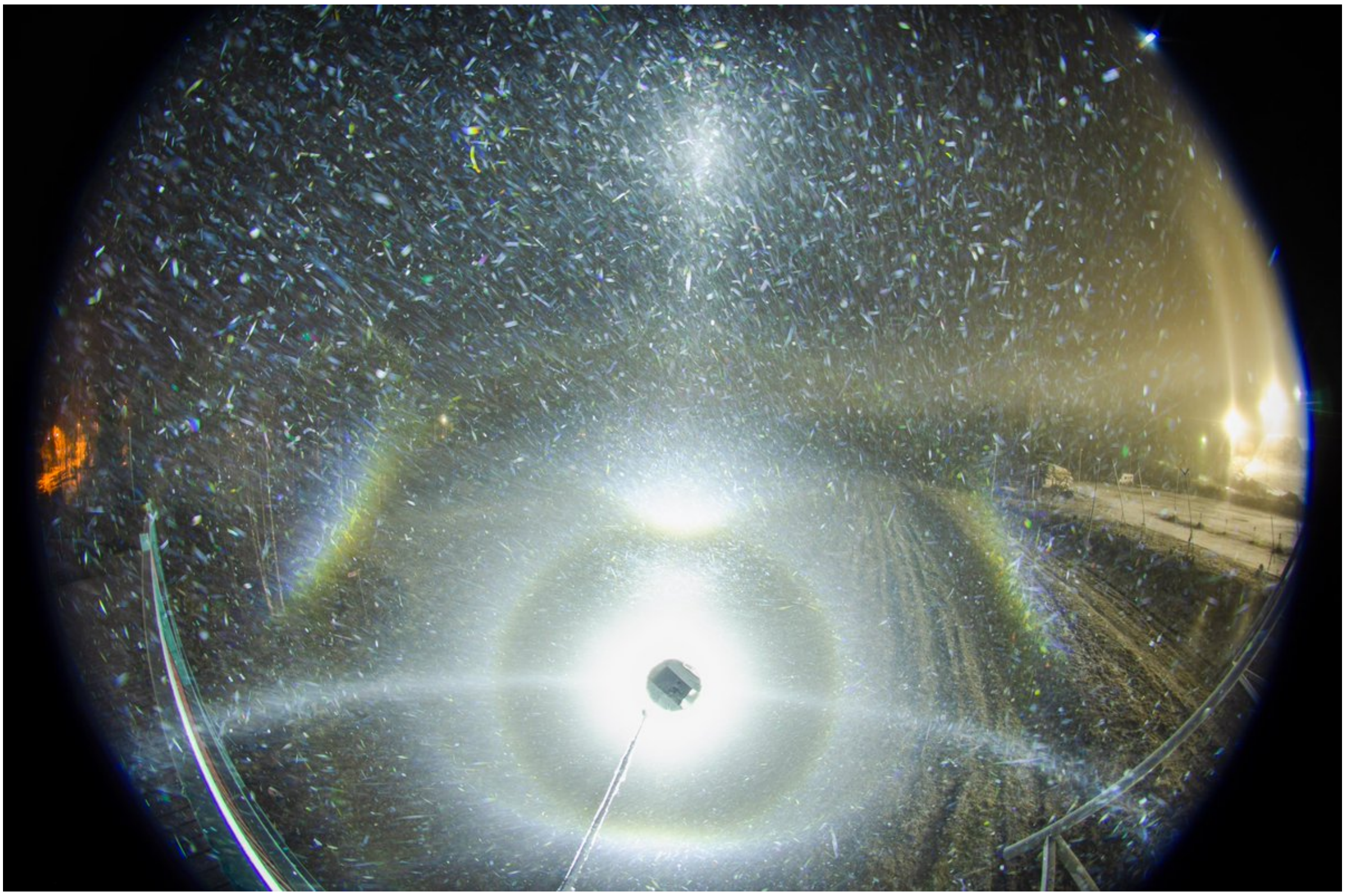


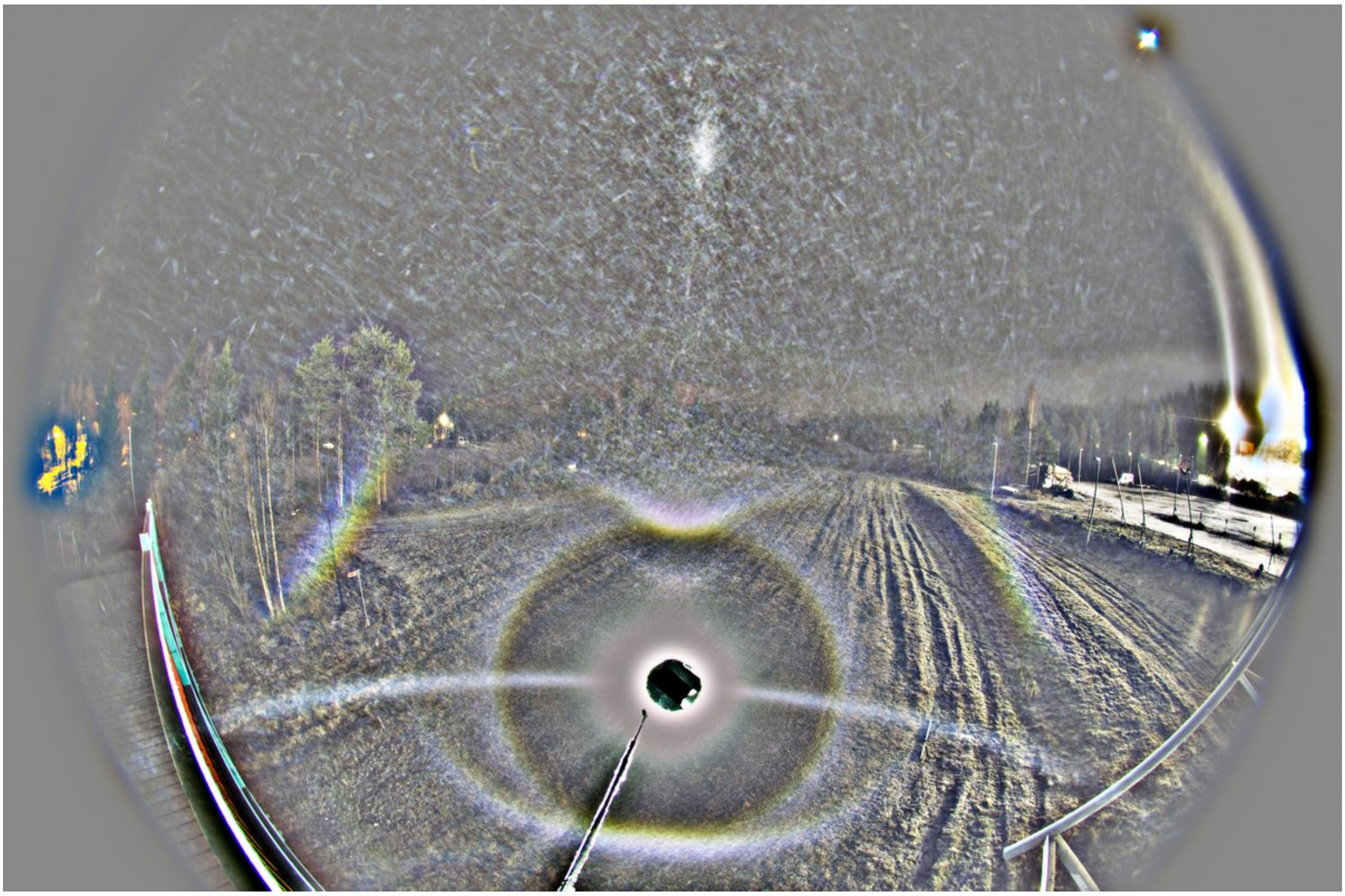
15/24 - 2025-12-04 13:25 UTC

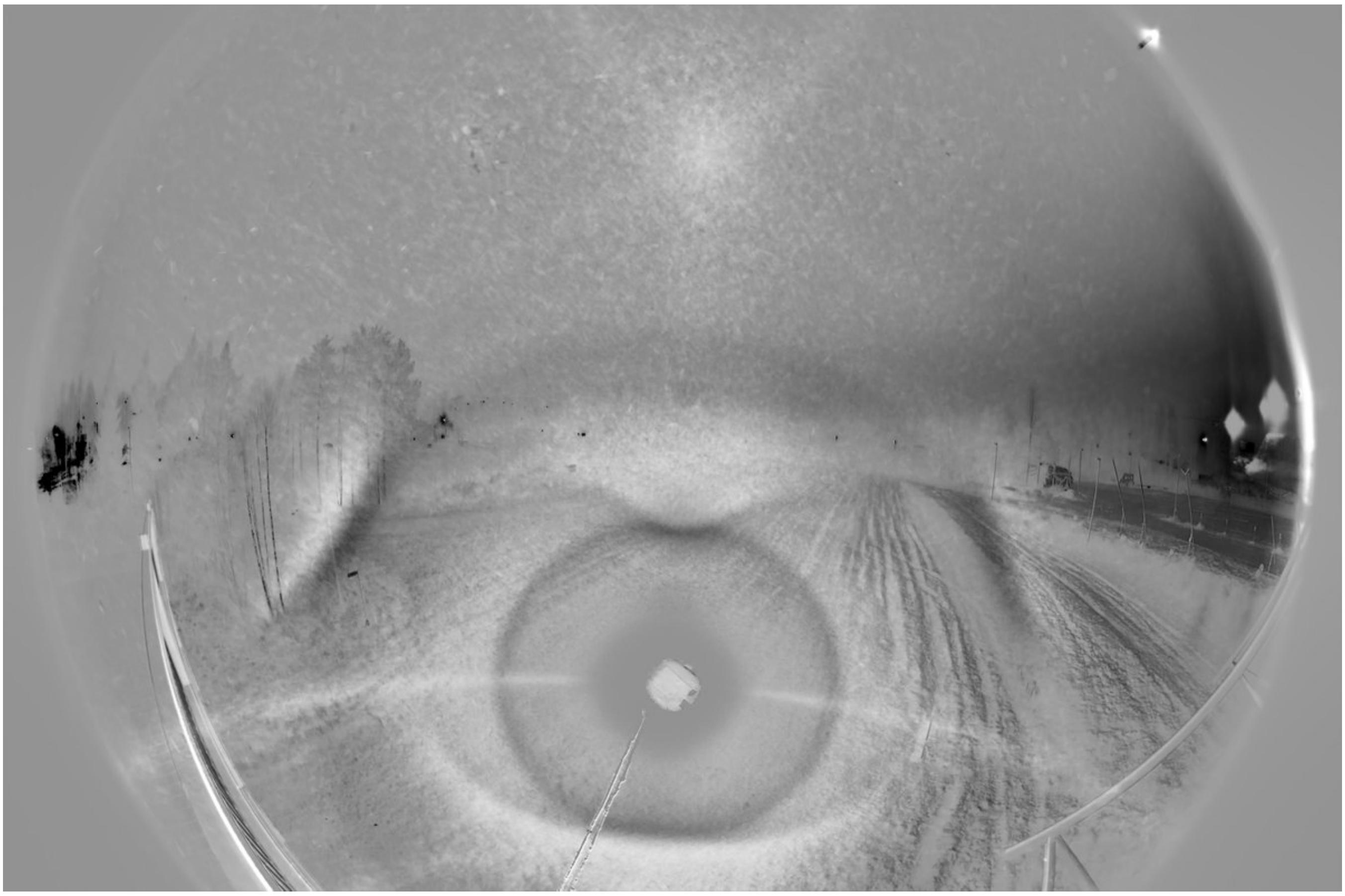
10/11 Nov 2025, Rovaniemi. Next Moila stack, light a little lower. Maybe around -35 degs.

By the way, I realized that the question "Is this the lowest elevation Moilanen arc?" above inherently supposes that mid-elevation is at 0°. Yes, light elevation terminology is truly in crisis. Kind of goes hand in hand with the crisis of halo nomenclature.

But something to rise for discussion of this specific stack. There is no parhelia, so possibly plates were no more at this stage. And flipsun is weak, as is expected from purely column swarm (there doesn't seem to be Parry here either). Thus it looks like Moila crystals don't have horizontal surfaces. This supports the idea that Moila crystal is a man of its own accord, not making other halos except within the Moila universe itself.







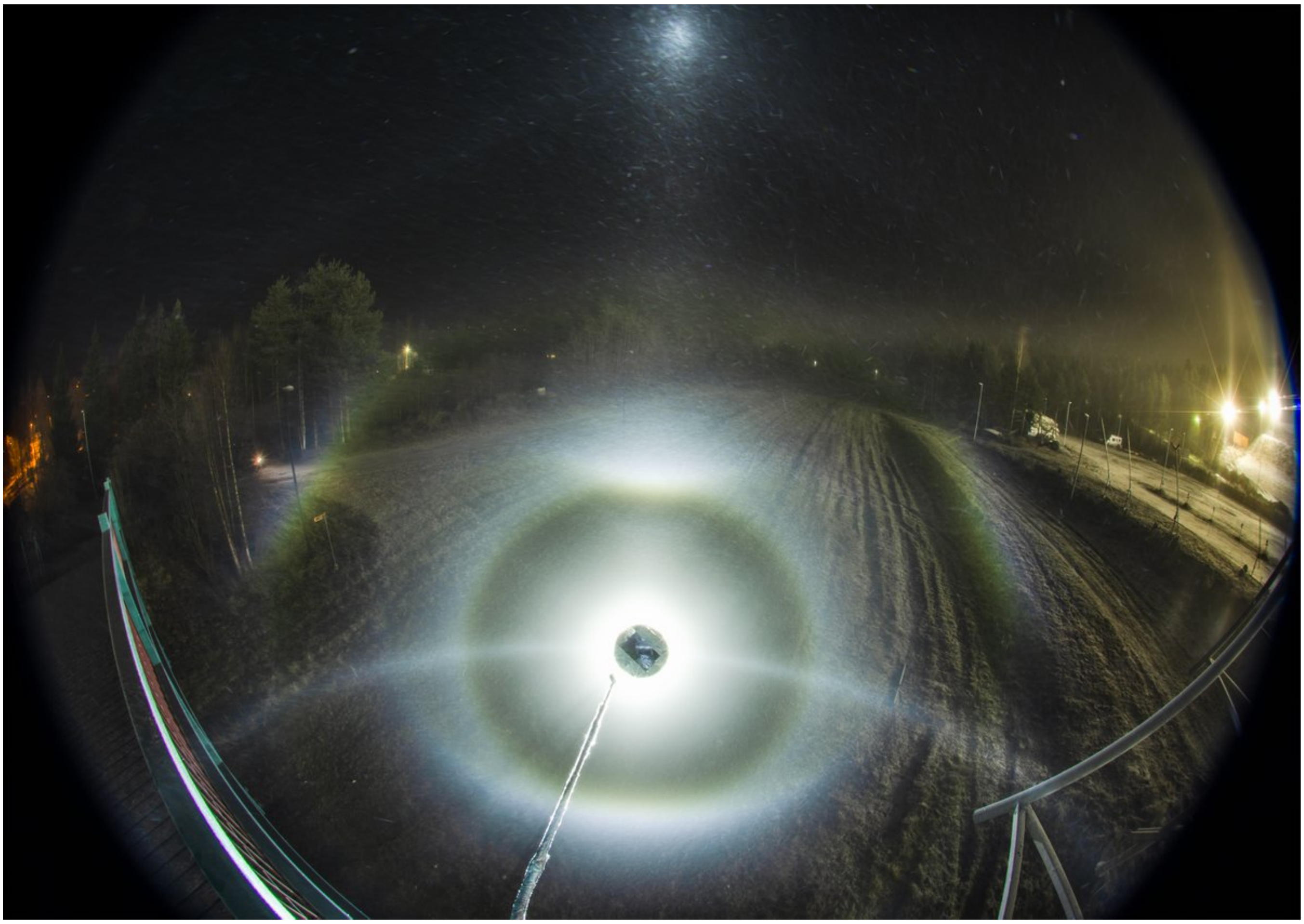
16/24 - 2025-12-04 14:10 UTC

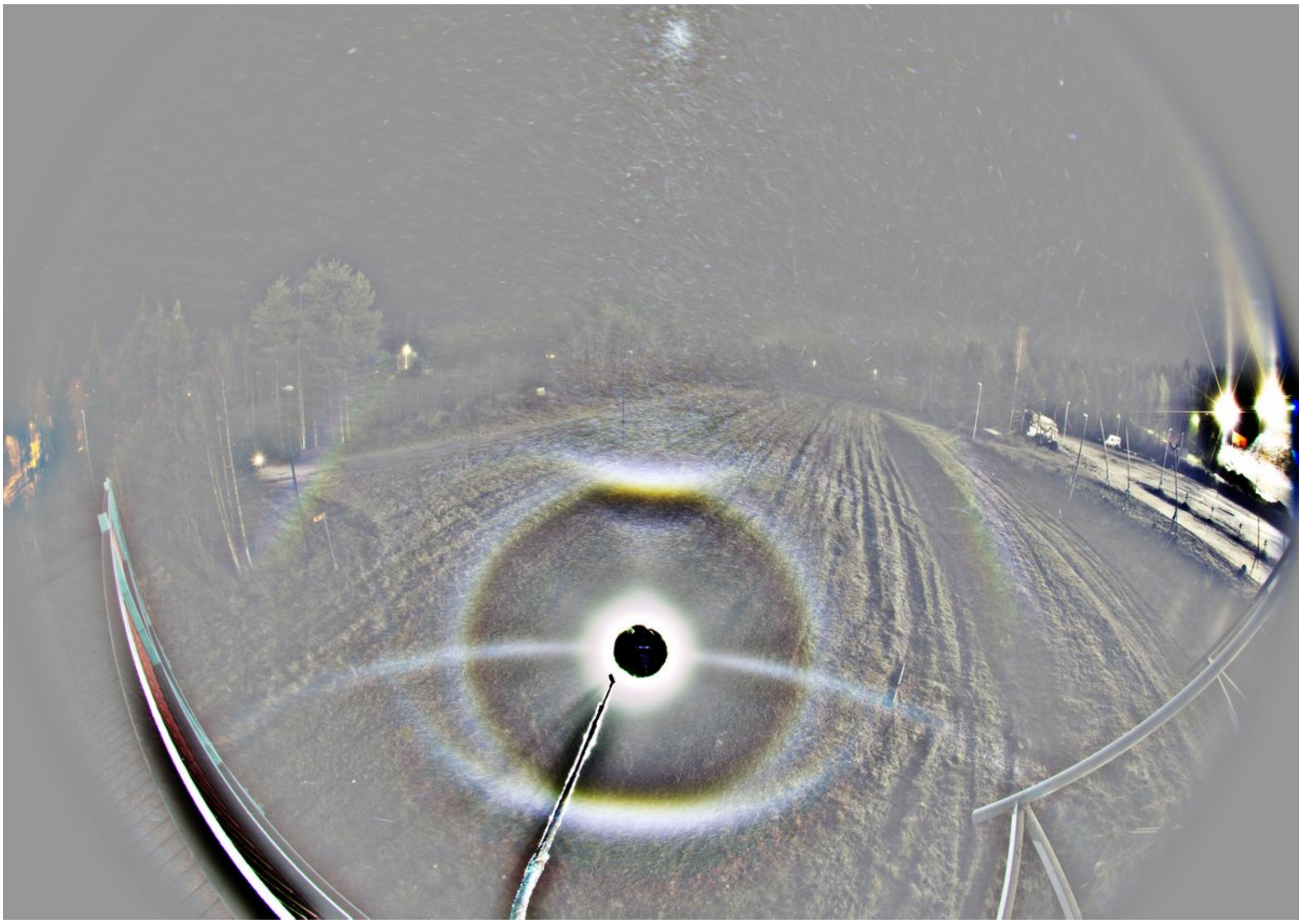
10/11 Nov 2025, Rovaniemi. Comparing the original stacks of the two previous post, where the difference in make-up of the displays is that the former contain parhelia whereas the latter is pure column. The flipsun brightness is indeed congruent with this change. That's an unadulterated column flipsun core (all column flipsuns are large, diffuse features, one of them making the 87° blue arc) there in the second image. And this comparison further bolsters the interpretation that Moila crystals don't contain horizontal surfaces. Stacks are 23 x 20s and 19 x 20s.

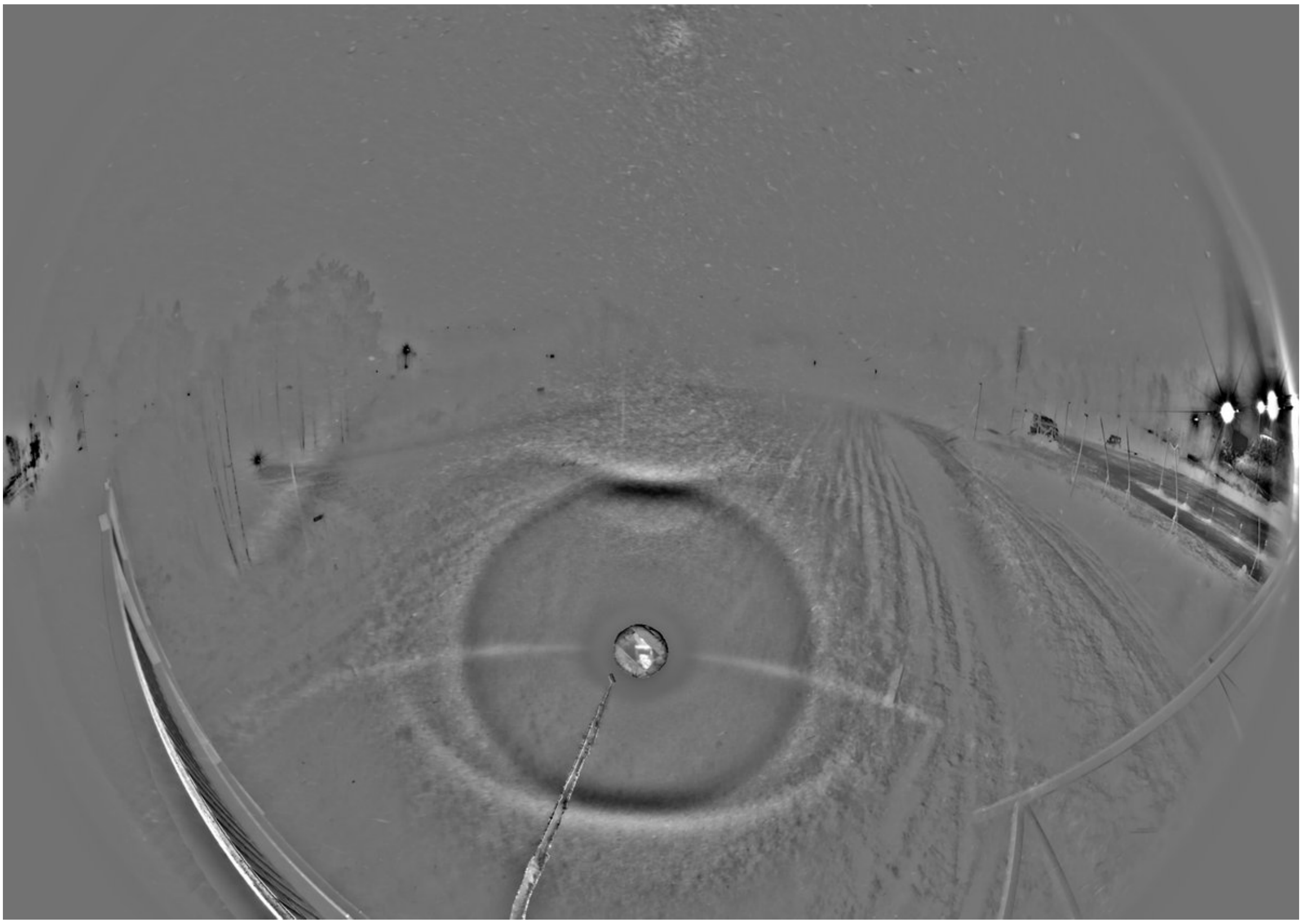
17/24 - 2025-12-04 15:05 UTC

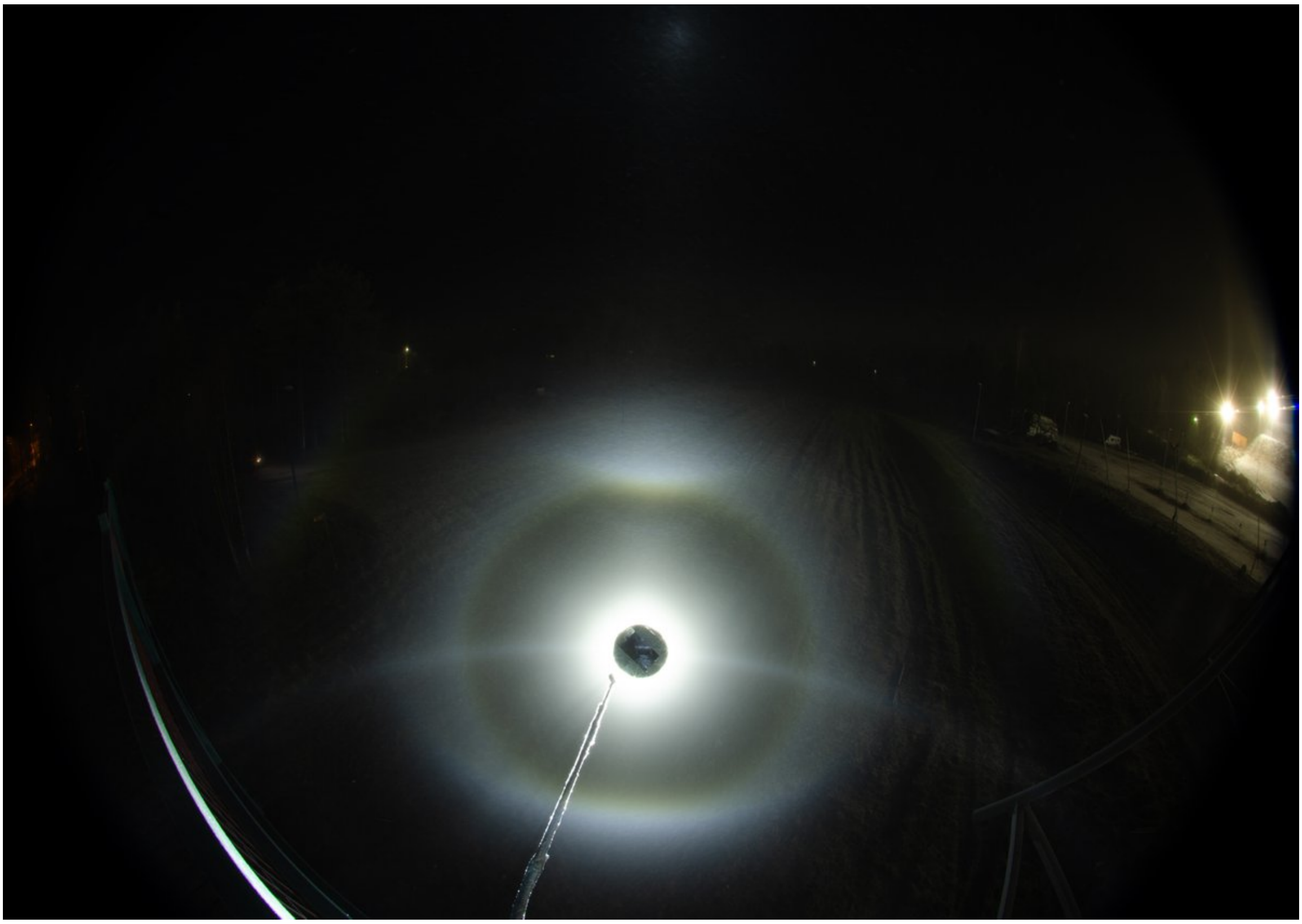
0/11 Nov 2025, Rovaniemi. Lowering the lamp further, I reckon now we are at least -40 degs. Moila has started to drift away from the lamp and looks like its seat is now occupied by the related halo from single oriented Moila crystals. So two Moila world halos here. Probably need to check simulations.

I thought I had more, but it appears that this wraps up the Moila elevation series. There is still plenty room for chasers to probe lower, for Moila's theoretical low end is at a little more than -70 degs. After this stack I still have one series with Canon for around 60°. That is when Moila is hovering close above the 22° halo like an inverted 23° plate arc, but the display had once again underwent an change and I think Moila crystals were gone. 25 x 20s, at 0501-0510.



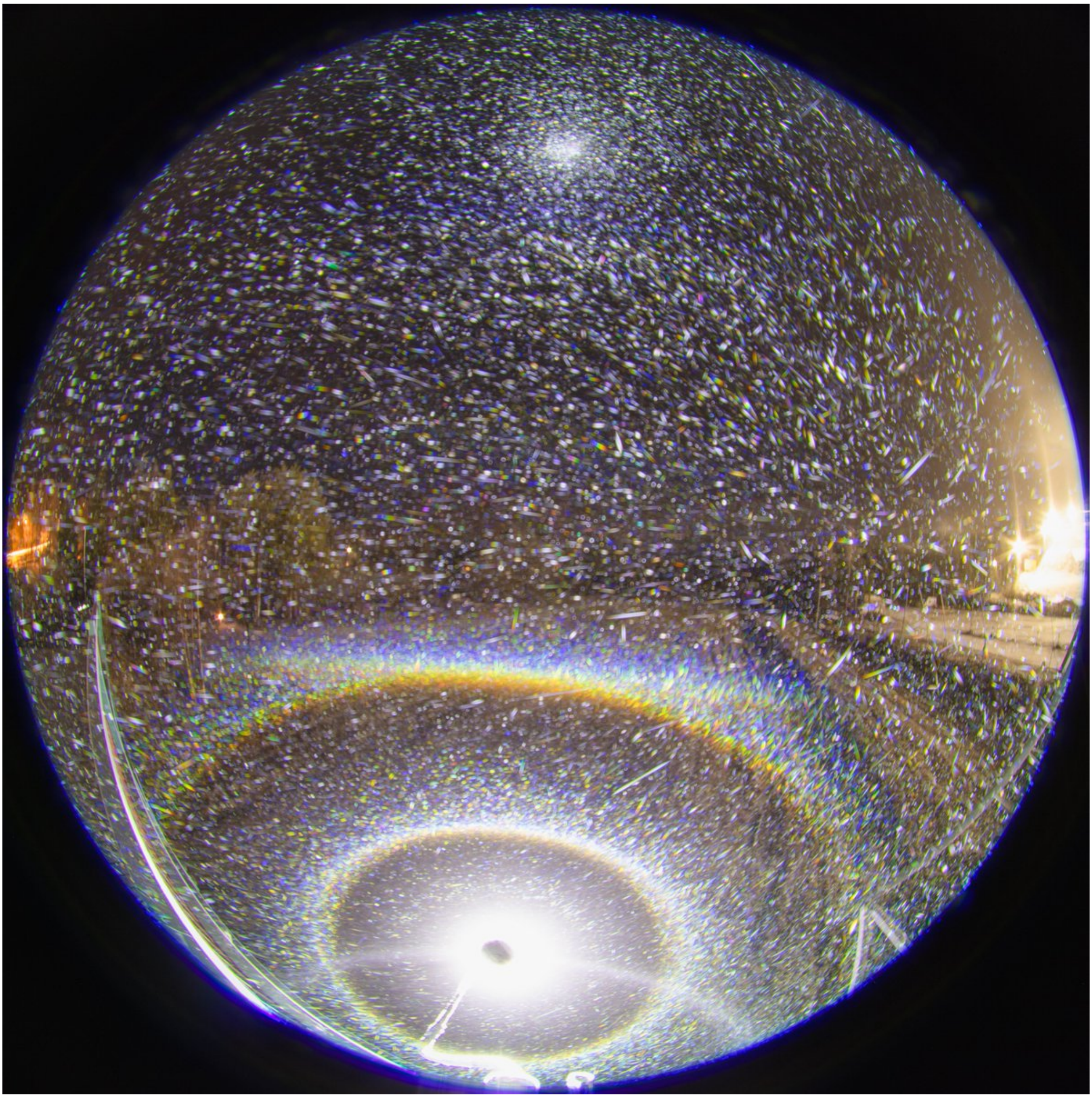






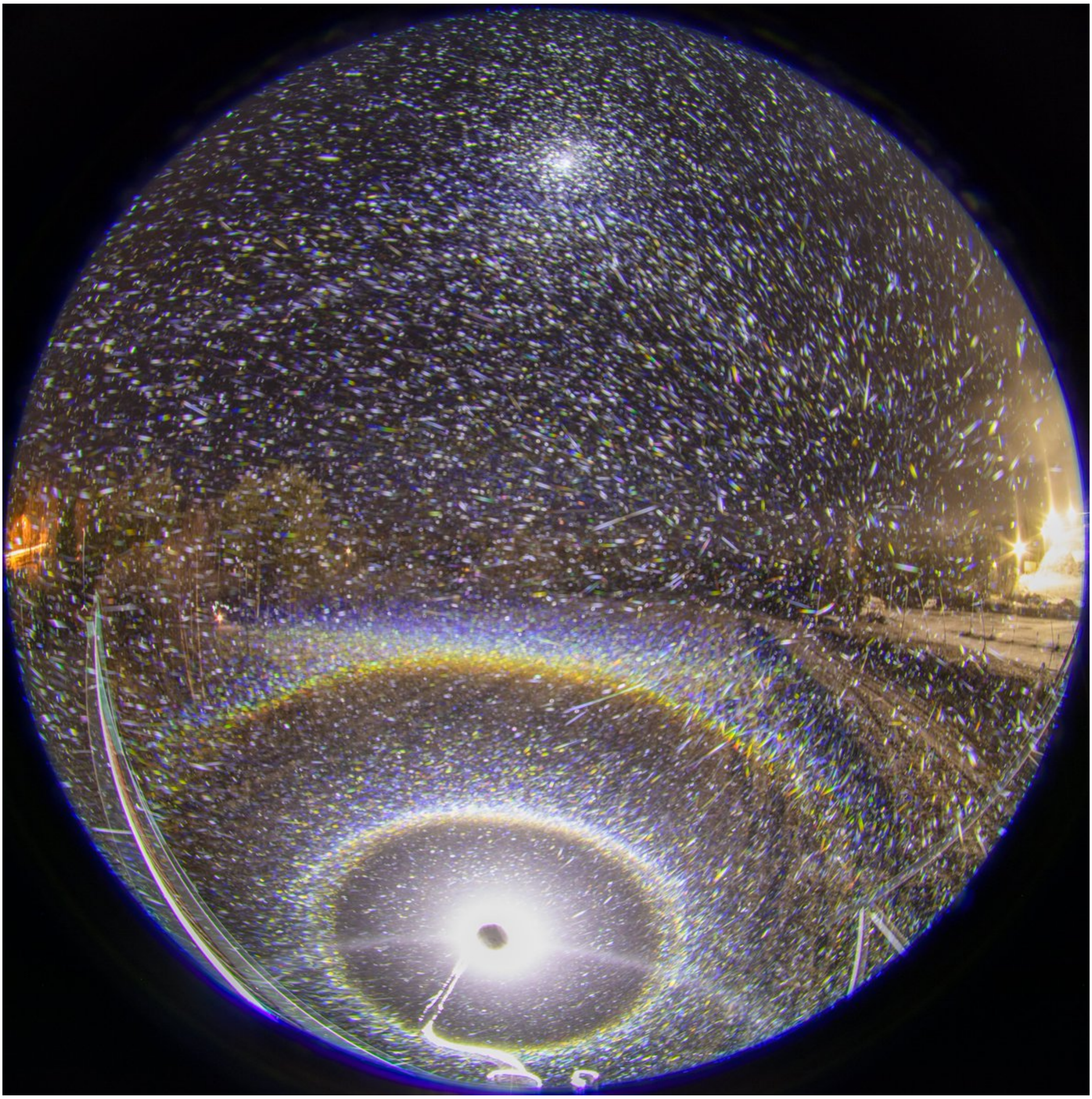
18/24 - 2025-12-04 16:07 UTC

10/11 Nov 2025, Rovaniemi. The second last set, it was nearing 6 am when the lights turn on the field below and on the bridge I am photographing. This stage had the best upper horizon arc and the flip upper horizon arc is back as well, visible above the horizon. No Moila above the 22° halo + tanarc (maybe there is some upper deepcave Parry, too, even if the separating tips can't be distinguished). 38 x 20s.



19/24 - 2025-12-04 16:29 UTC

10/11 Nov 2025, Rovaniemi. The very last set. I moved the lamp a bit closer to the horizon. I was aiming for around -58 degs, which is about the limit elevation for horizon arc, but didn't succeed – this is maybe around -60. Anyway, I managed 13 frames before the lights came on. It looks like the display was improving here, the tip of Parry arc is separating on the left side even if this stack is 1/3 of the previous. The flip upper horizon arc is there just like in the previous stack. It is seen well even in single images.



20/24 - 2025-12-04 18:21 UTC

10/11 Nov 2025, Rovaniemi. And this is how was ended by the lights at 6 am.

[video: 1996646033196306866-jU0w-ikgMAQpqnOx.mp4]

21/24 - 2025-12-05 07:30 UTC

10/11 Nov 2025, Rovaniemi. After the lights got on I took this quick photo of a zenith arc from ski center lights. Selfie not for vanity but purely practical reasons to block the horrid bridge light. Olight Javelot Turbo was still burning down on the ground, responsible for the beam coming from behind my shoulder.

22/24 - 2025-12-06 01:45 UTC

10/11 Nov 2025, Rovaniemi. Daylight was already giving signs of itself at the horizon and the last thing I did was to take these photos under an end-of-road streetlight at the ski center. The plate halos – parhelia, parhelic circle, zenith arc and horizon arc – were magnificent 3D surfaces. A top experience. But regarding photography, it was TLHS, the intensity having dropped to like half what it was just a moment ago.

Anyway, I was looking also for 120° parhelia, but was not able to see it. Some arcs extend obliquely down from the lamp, particularly in the first photo, but is there also an obliquely up component? If so, is this rather helic arc than 120° parhelion? But of course a display this extreme should have had the 120°.

The white narrow arc I interpret as lunar parhelic circle. At 7 am when the I took the photos, 78% moon was at 48 degree elevation. Didn't notice it visually.

But what is that colored circle that delineates the dark void high up around zenith? This springs to mind the display by Luomanen on 29 October 2014 https://t.co/Vixr831kxo I have this memory that at the time Tape explained the void as an 3D effect from snow reflection of the halo torch light and there is a discussion in comments somewhere. But it is entirely possible that I am hallucinating the whole thing.

There is a handful of other spotlight displays that suggest similar, though much weaker void. Such as https://t.co/noOUwBQXyy https://t.co/iNLHq6x8Qu https://t.co/OZSj4rXqlh

While I may have imagined Tape's role, I found Lefadeux's comment on the same issue in Eresmaa's streetlight display observation: https://t.co/SUilb2PtDy He regards an arc that was initially taken as zenith arc from distant lights "integrated divergent pillar". If indeed this is seen also in my photos, I could swear I see red color. And that throws spanner in the works because a blue edge effect is supposed to not have red (see also the next post's b-r versions).

An integrated flip-nadir arc circle + its extension from snow reflection would appear at the same 58 degree elevation from the horizon as the all-around blue flipsun, and would allow for red. However, there is a snag that the intensity threshold would be the other way around, just like Lefa says in his comment about Wegener. But as both raypaths must clearly exist here, could it be that this is an amalgam of both? I suppose the flip nadir arc mechanism would be much fainter than the flip-sun mechanism, so I have doubts if the red could be seen at all.

But yet if somehow the red would actually be due to flip nadir arc mechanism, then the observation would have some bearing on the "integrated zenith arc" seen in the displays of Jutland, Siziwang Qi and Levi. Actually, Ji Yun has alternatively visioned a reflection from snow as explanation. This is info is relayed by Jia Hao in his post Hidden Treasures in the Jutland Display https://t.co/Vckjx7cLiR Another Jia Hao's post, The post Secondary CZA from Sun Pillar, is also relevant here: https://t.co/FwgEYlKzWy

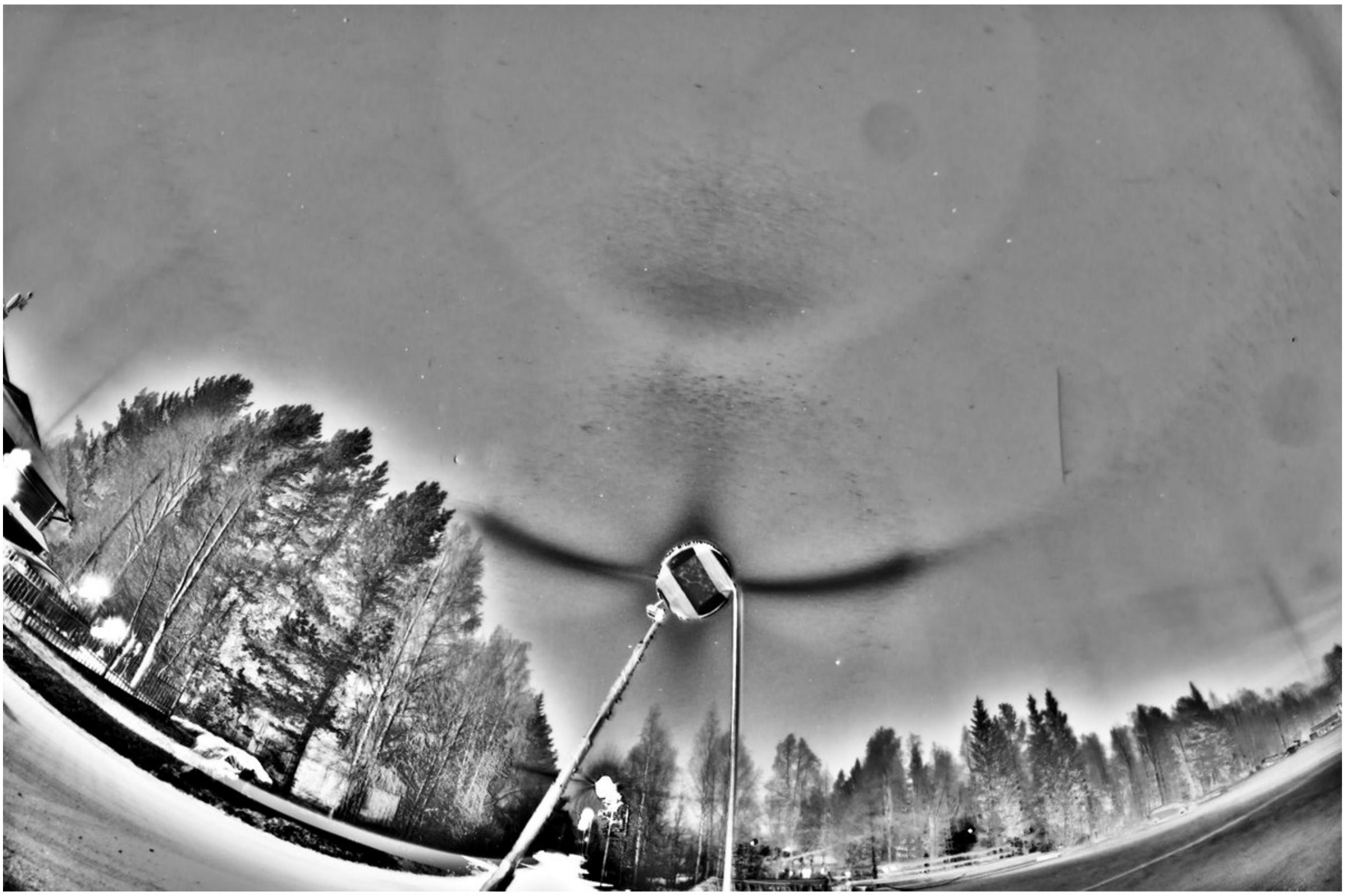
---

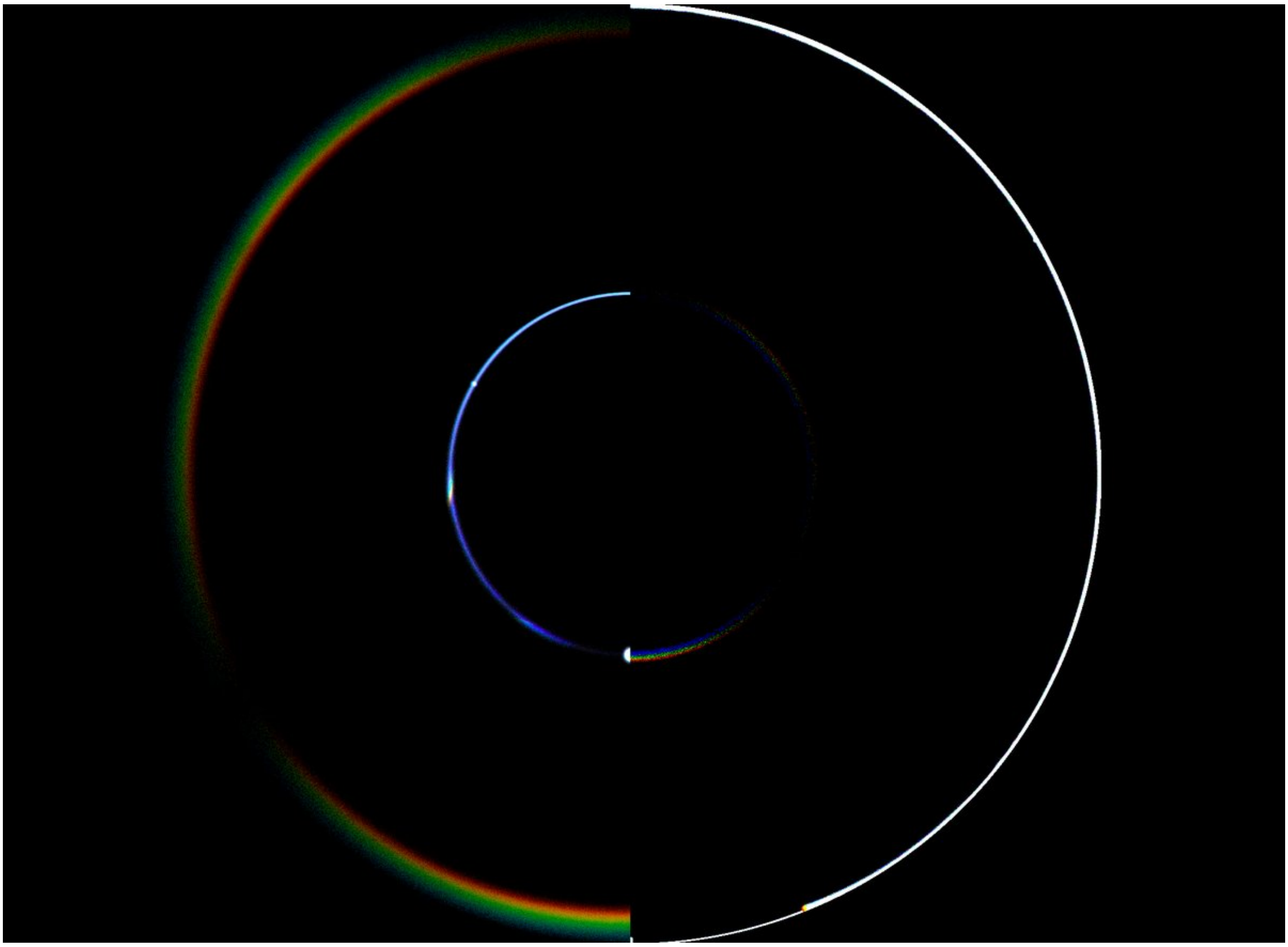
23/24 - 2025-12-06 02:11 UTC

10/11 Nov 2025, Rovaniemi. b-r of the previous and a simulation showing how the flip parhelic circle blue edge effect and flip-nadir arc circle + its extension both appear at 58.5 degs elevation.

All right folks. It's a wrap concerning X. I just realized today the site is hiding a big chunk of my earlier posts. I can't see anything prior 24 January 2024 when scrolling down. If I use advanced search by giving the date range, I get 13 days more, to 11 January. The data is there, but I can only access it by asking Grok. Grok confirmed that this is a common glitch of X.

I'll look what other options there are. I guess all these sites are the same. While some site might look like a suitable option now, how it will be in 10 years. Grok suggested Bluesky and Mastodon, but what I had a quick look, they are like X.





24/24 - 2026-04-20 13:23 UTC

10/11 Nov 2025, Rovaniemi. In the post two up, a glitch made the images vanish, so I am reposting now. I also happened to find one more view, the portrait orientation in the three image collage. This view constituted the last photos I took, the display was vanishing and is markedly weaker than in the two landscape orientation photos. But it helps in building a picture of what's going on.

In the portrait image, the strong arc above the moon / opposite to the streetlight is the zenith arc from the ski center lights. But it is probably also counterzenith arc, as for crystals higher up many of those lights would be below horizon (they are not downwards-only lights like streetlamps, so they would light up good also higher up crystals).

But what about the associated 58° circle in landscape photos? Could it be (counter)zenith arc from other lights in the surroundings? However, the other lights outside the ski center are just normal streetlights, the shade largely blocking upwards light. So I am rather leaning on the 58° circle being a (counter-)Dufay from the ski center lights.

Again, I refer to the December 2013 Koli display where we were facing somewhat related issues:
https://t.co/psIypmy8AQ
https://t.co/i2XqMRp7lP

It is worth noting in the Rovaniemi display that the 58° circle in the landscape photos has enhanced blue component in b-r opposite opposite to the ski center lights / above the streetlight. Is this the integrated countersun blue edge from ground reflection of the streetlight? At least in this plate display the feature was strong: https://t.co/knI5PMzOyO

But the integrated countersun should really be lighting up the whole 58° circle, superposed with the other arc. And I think the dark void at the center of the circle indeed tells that it is lighting up the whole circle, because that void can only be created by the integrated countersun.

So in conclusion, there's two types of halos superposed here: the blue edge from simple internal reflection, and the circle from 90 prism refraction (the exact type undetermined).

Or was this all so much of baloney? In any case, in such restless light environment as here it is good to have another camera snapping photos some distance apart from the streetlight to allow to subtract the streetlight component from the situation. And as Lefaudeux suggested in a mail to me, a drone would come handy to see what are light sources that the crystals higher up see. That's already aplenty for man one to do, not to talk about if crystals are also to be sampled. In halo chase, much more could be achieved with an organized team.